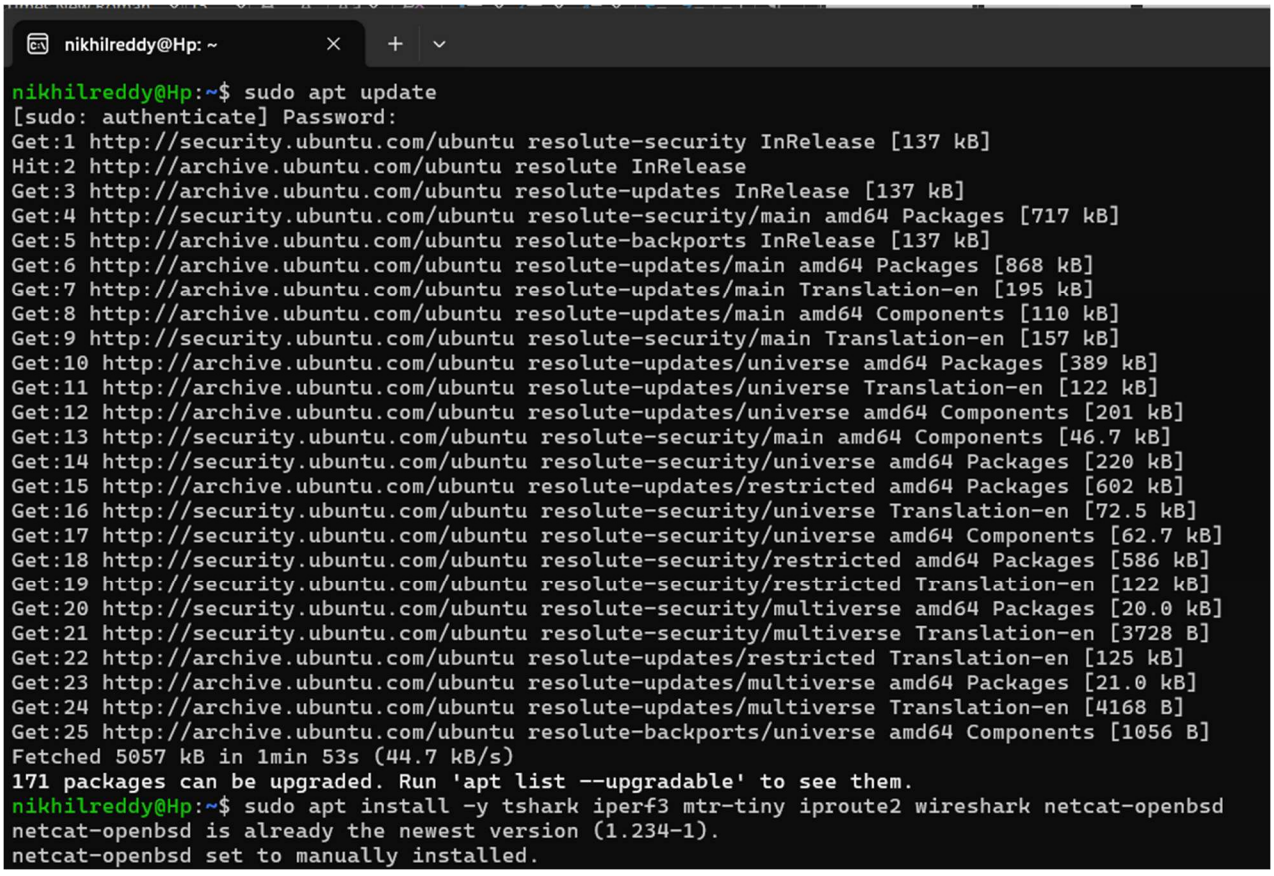
CNNP- Skill 2

THREE-POINT CAPTURE – Analyze packet captures from three points in a network path to understand packet flow and identify performance differences.

STAGE 1 — Install tools

Open Ubuntu and type the following commands in the Ubuntu terminal

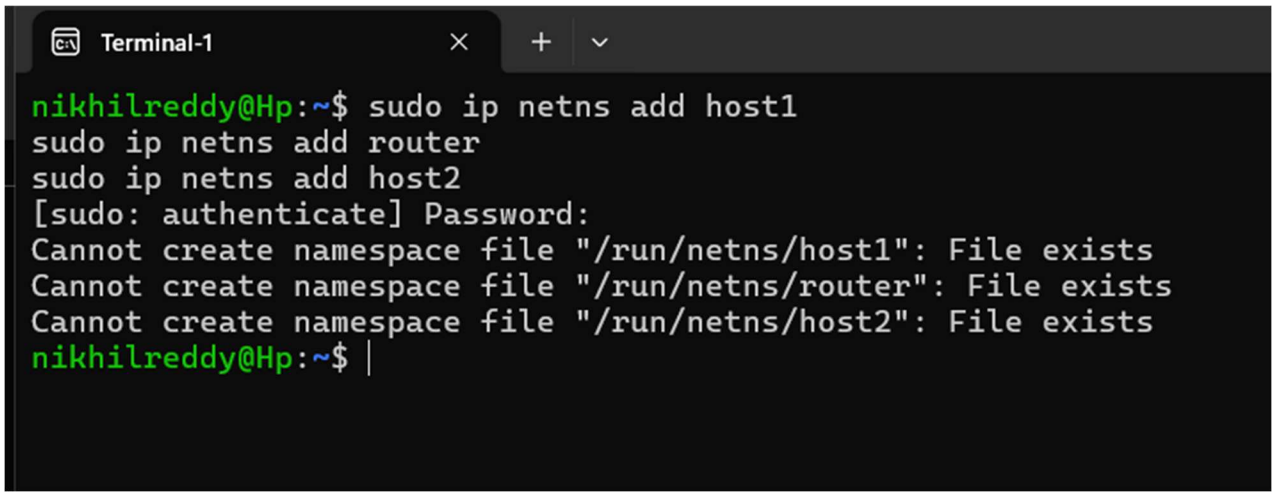
sudo apt update sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd



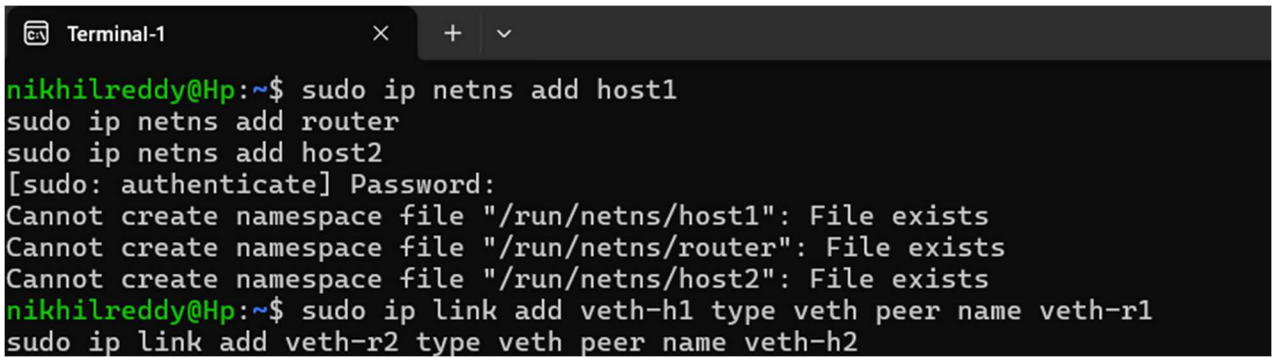
STAGE 2 — Build the network

Open again ubuntu and rename that tab as Terminal 1 and run all of these one by one:

sudo ip netns add host1 sudo
ip netns add router sudo ip
netns add host2

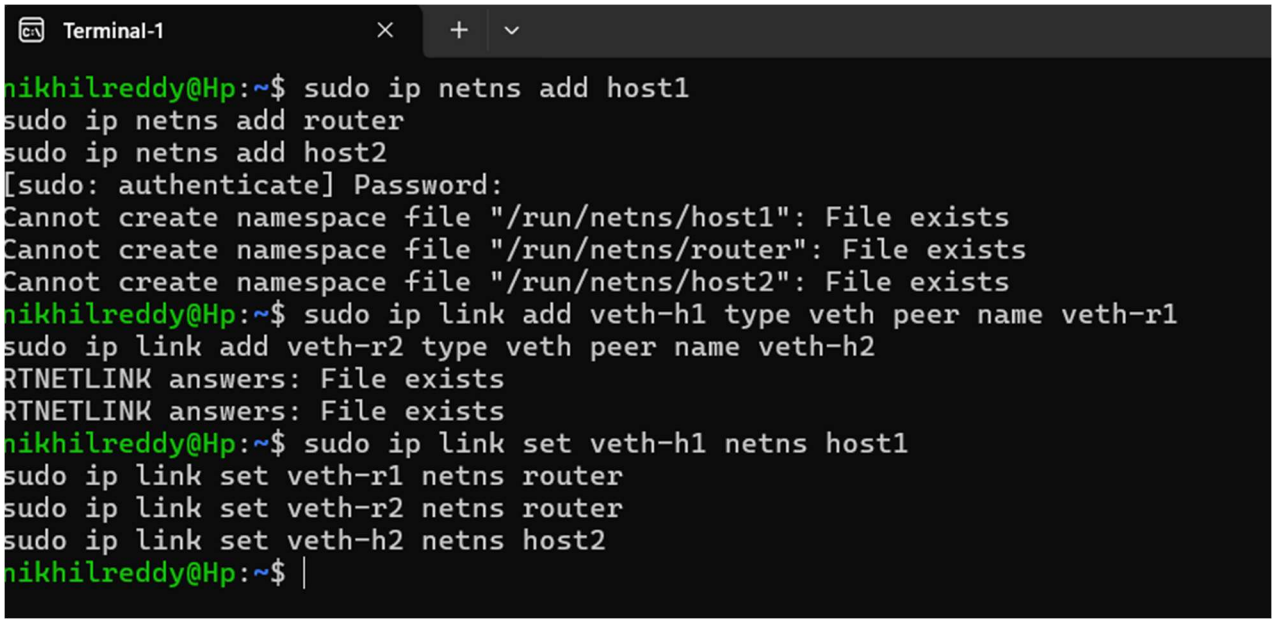


sudo ip link add veth-h1 type veth peer name veth-r1 sudo
ip link add veth-r2 type veth peer name veth-h2



sudo ip link set veth-h1 netns host1 sudo
ip link set veth-r1 netns router sudo ip

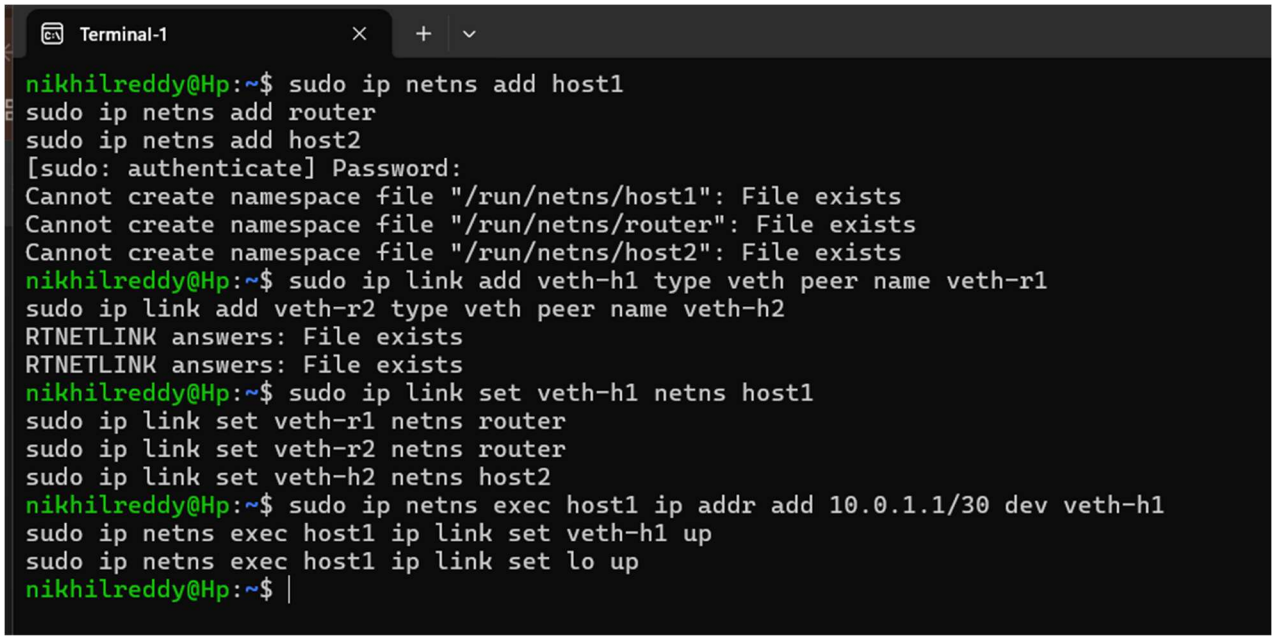
link set veth-r2 netns router sudo ip link
set veth-h2 netns host2



STAGE 3 — Configure IP addresses (Note: In terminal 1 Only)

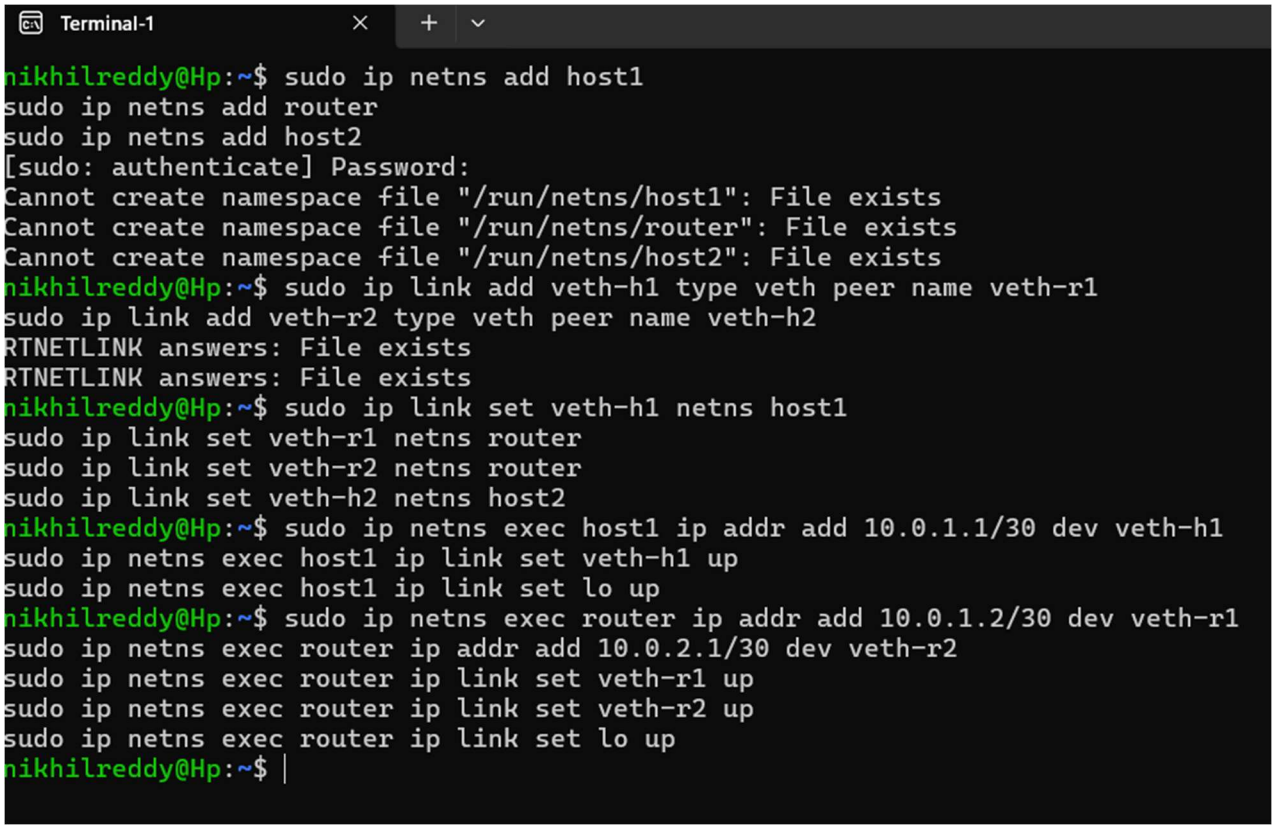
host1:

sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up sudo ip netns
exec host1 ip link set lo up



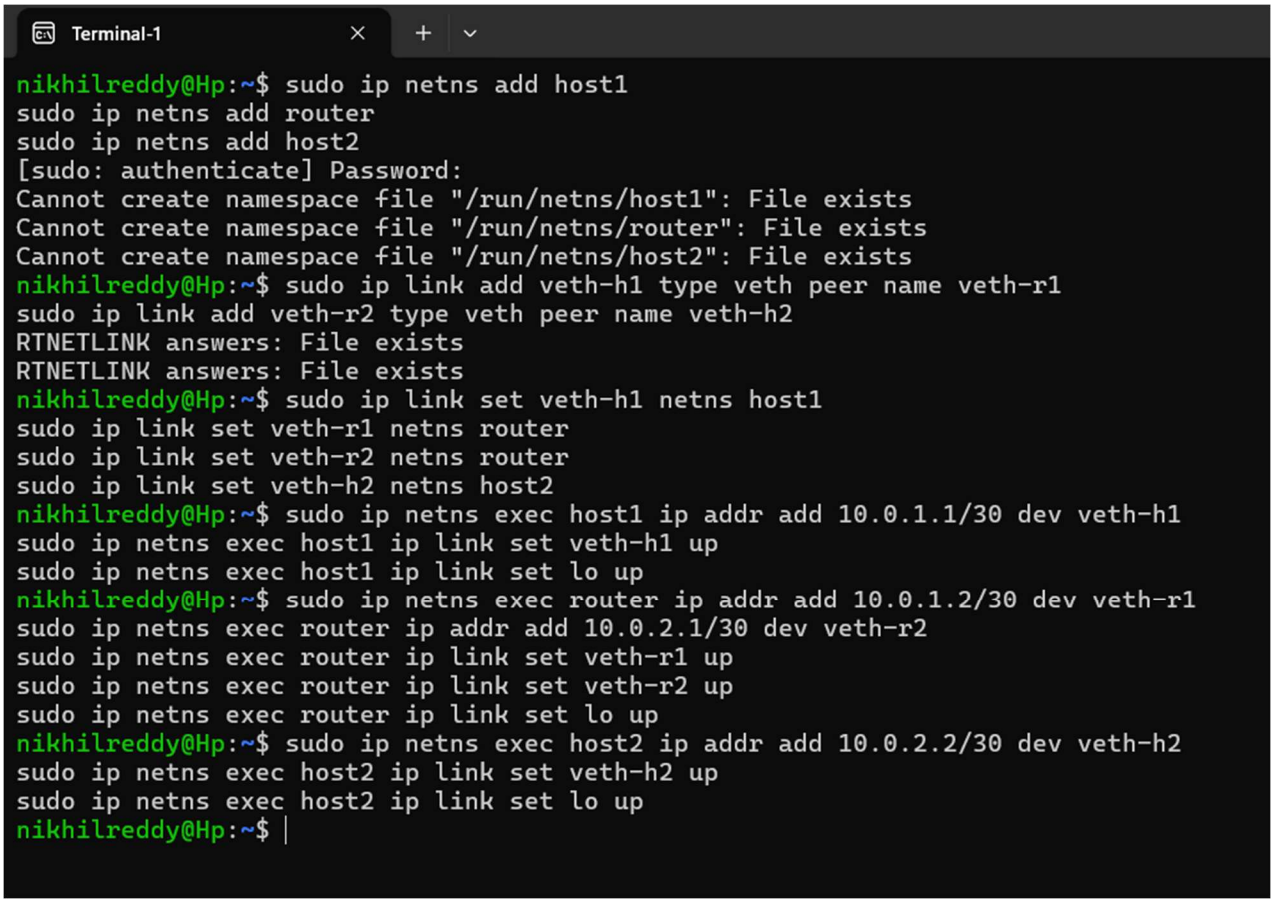
router:

sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up sudo ip netns exec router ip link set veth-r2 up sudo ip netns exec router ip link set lo up



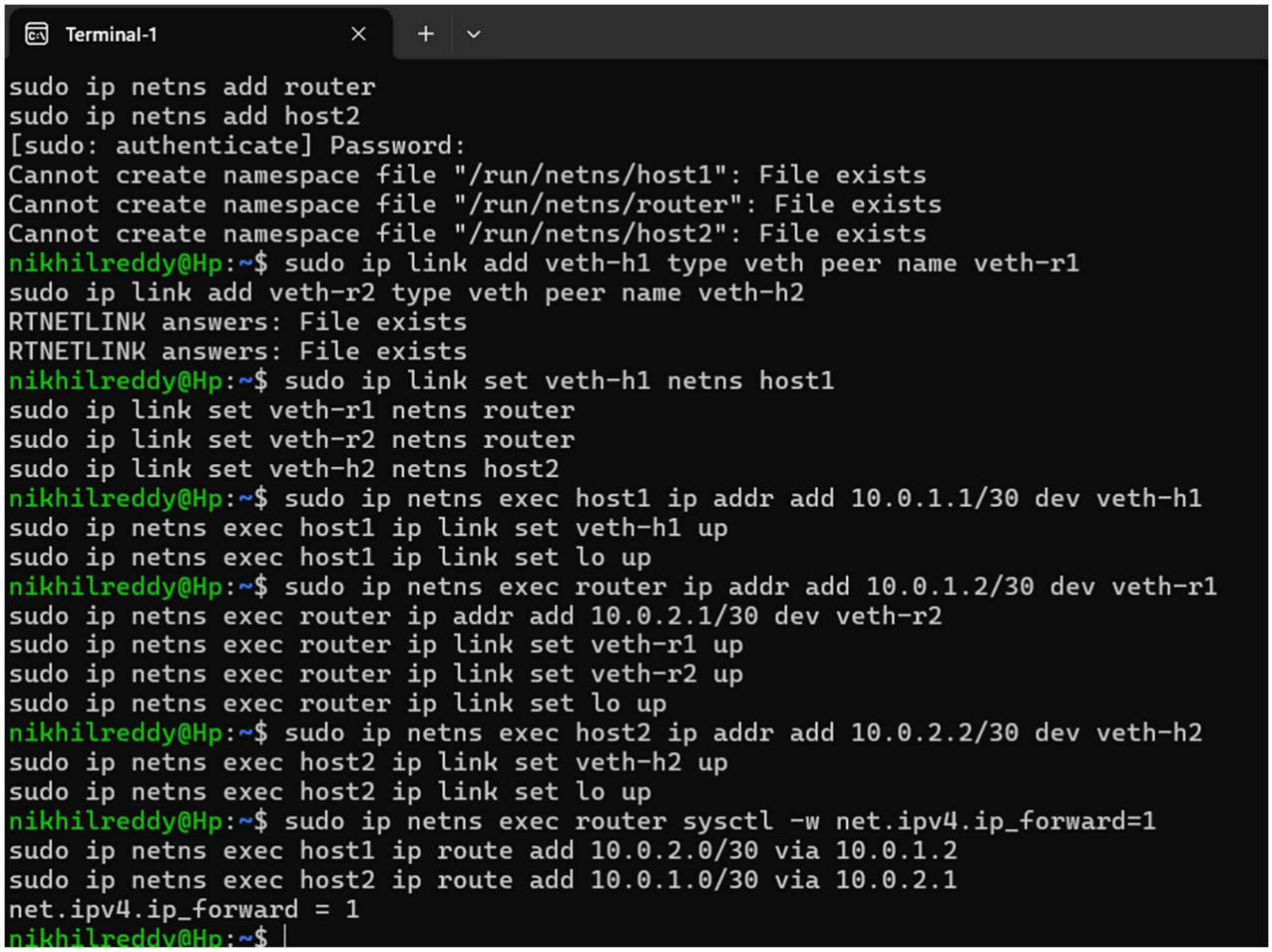
host2:

sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up sudo ip netns
exec host2 ip link set lo up



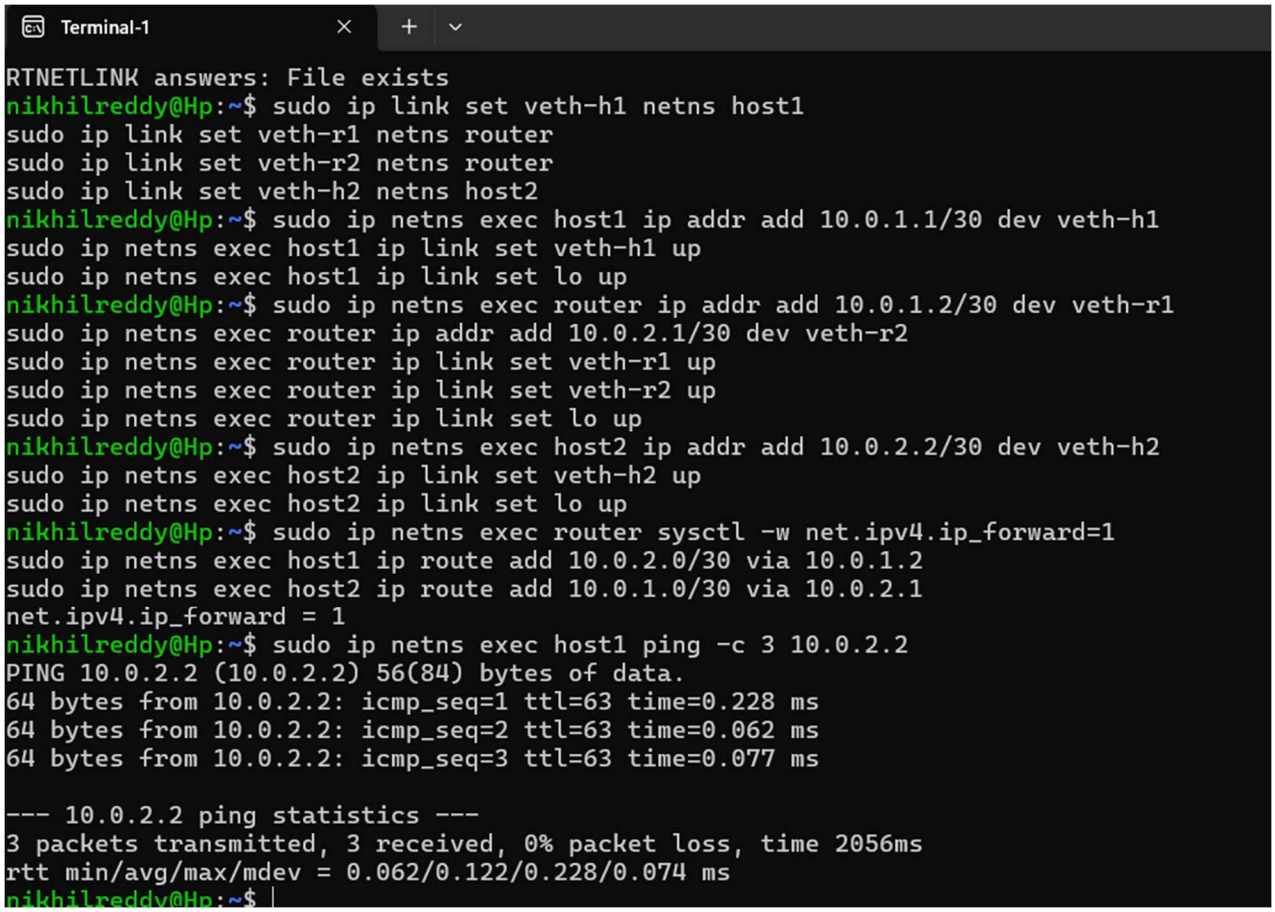
STAGE 4 — Enable routing

sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1 sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2 sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1



STAGE 5 — Test connectivity

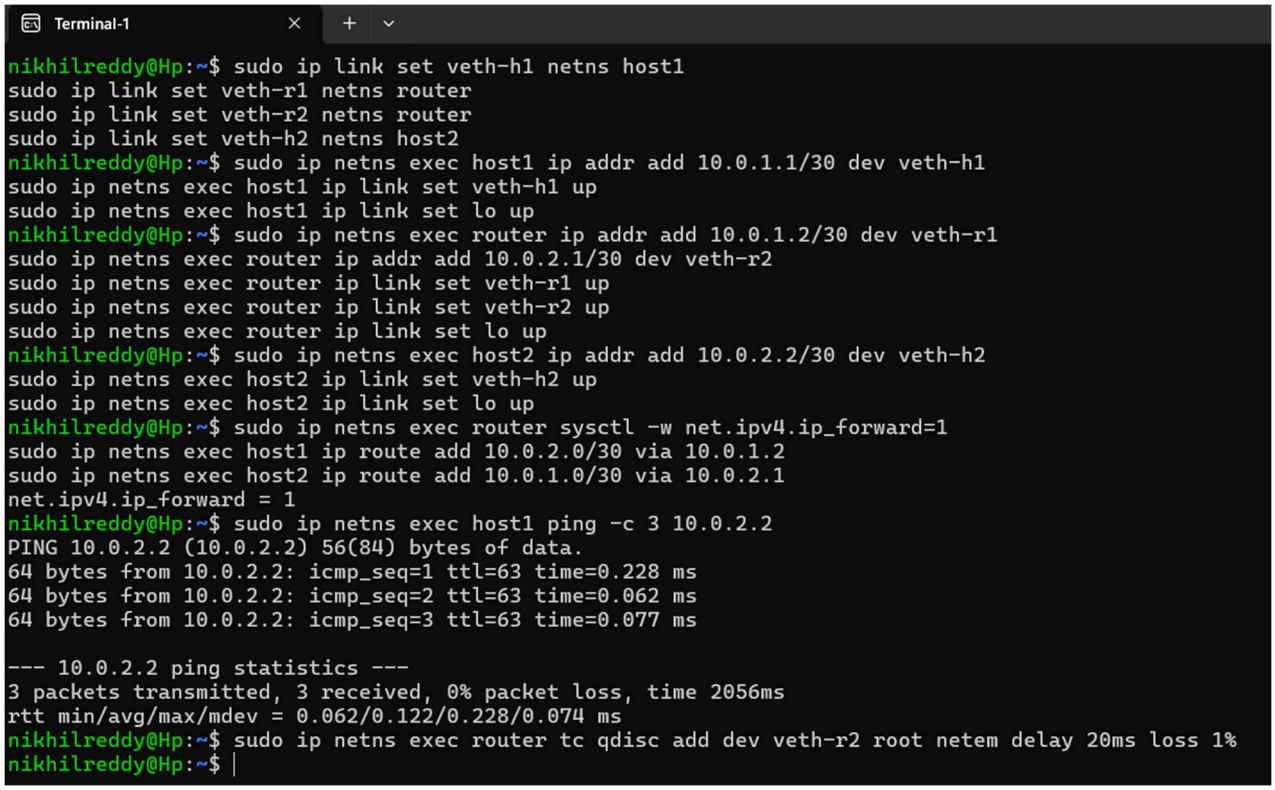
sudo ip netns exec host1 ping -c 3 10.0.2.2



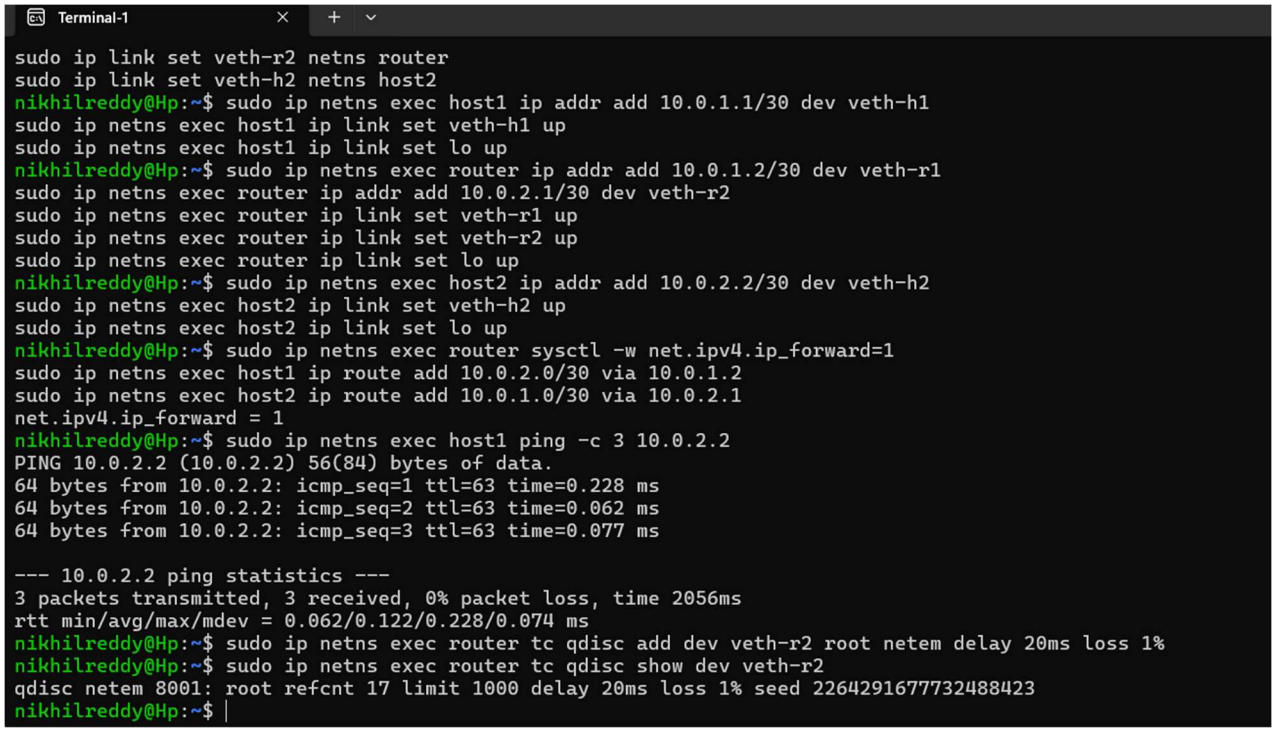
You must see:

3 packets transmitted, 3 received, 0% packet loss

STAGE 6 — Add delay and loss



Verify: sudo ip netns exec router tc qdisc show dev veth-r2

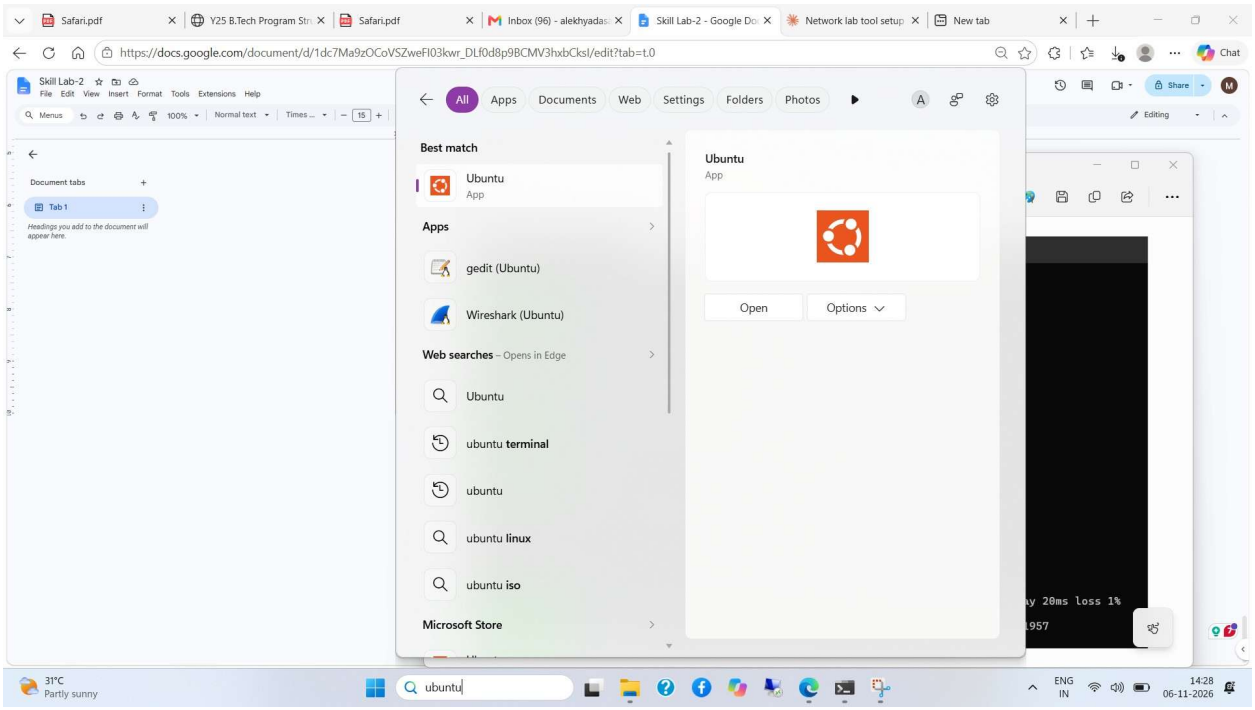


You must see netem delay 20ms loss 1%

STAGE 7 — Open 4 terminals

You need 4 terminals open now.

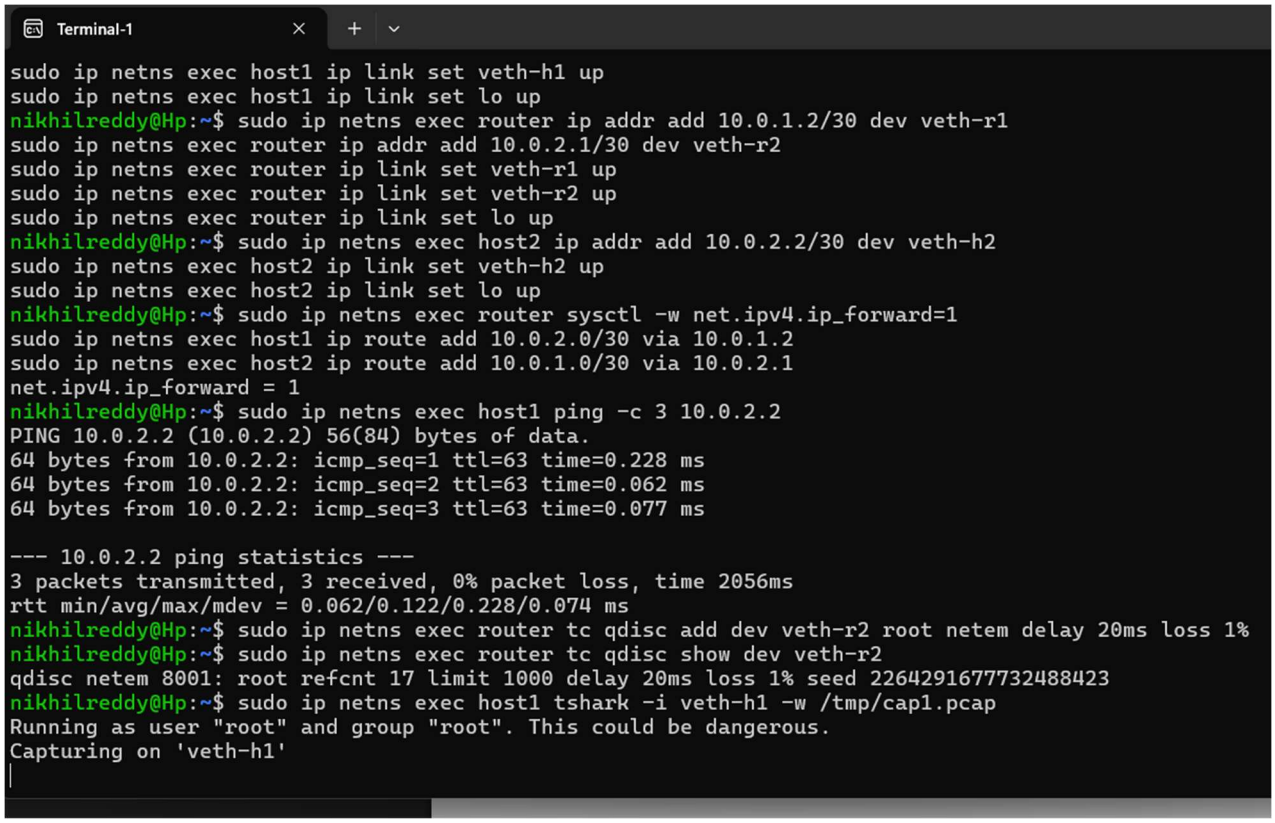
Terminal 1 — already open (used above)
Terminal 2 — open new (By going to Ubuntu option)



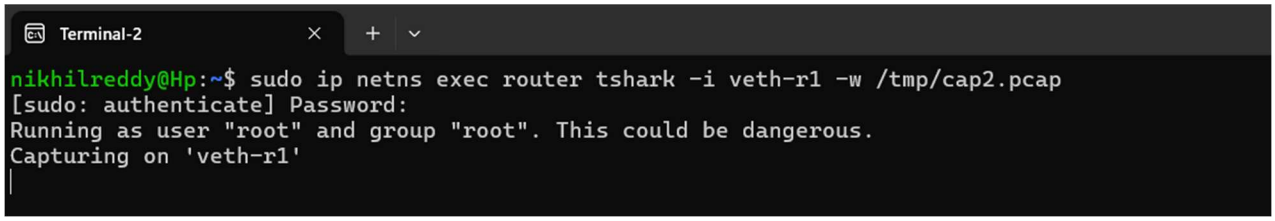
Terminal 3 — open new (By going to Ubuntu option)

Terminal 4 — open new (By going to Ubuntu option)

STAGE 8 — Start three captures simultaneously Terminal 1 — capture at host1 sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap

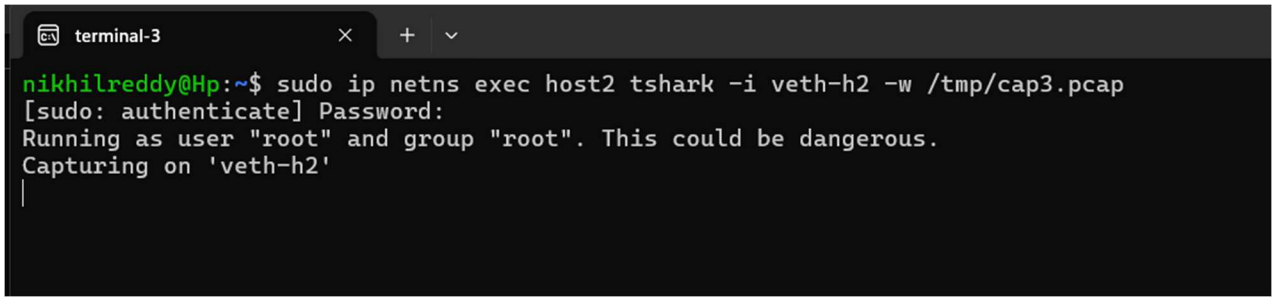


Terminal 2 — capture at router sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap



Terminal 3 — capture at host2

sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap



All three show:

Capturing on 'veth-h1'

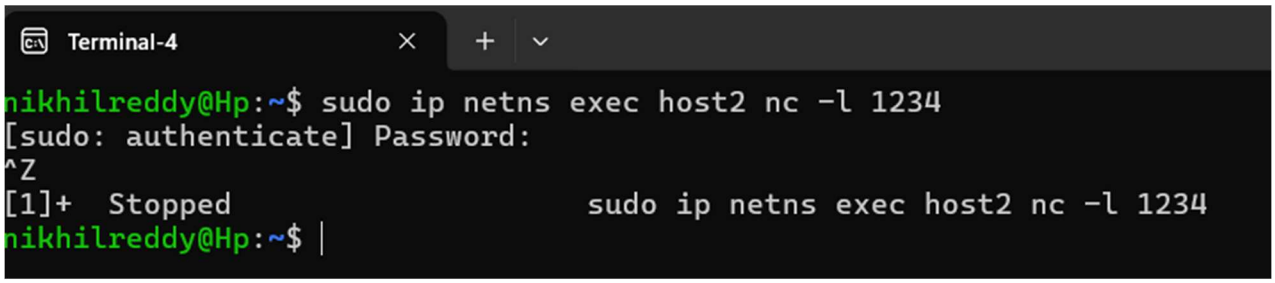
Leave all three running.

STAGE 9 — Send real message from host1 to host2

Go to Terminal 4.

First start the receiver on host2:

sudo ip netns exec host2 nc -l 1234



Open a **5th terminal.**
Connect from host1:

sudo ip netns exec host1 nc 10.0.2.2 1234

Now type this message and press Enter:

Hello from host1!

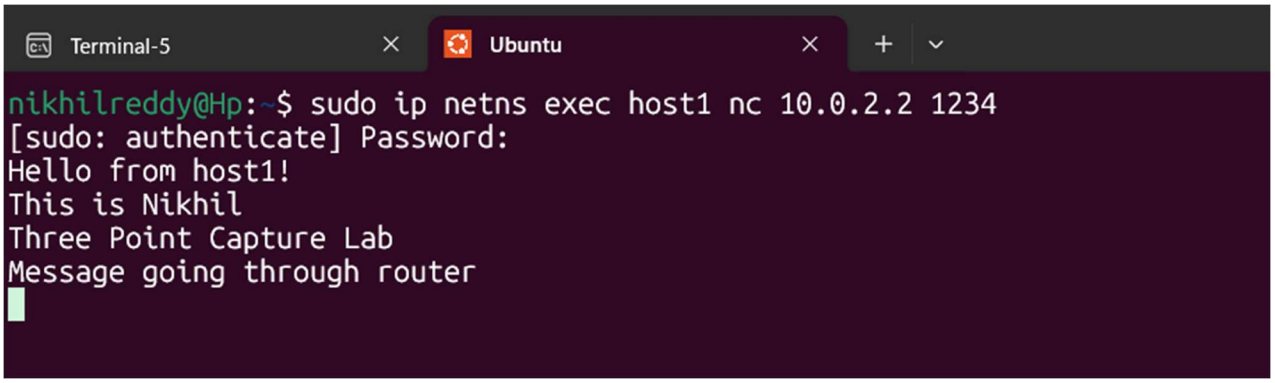
Type more messages:

This is Alekhya

Computer Networks Lab

Three Point Capture Experiment

Message traveling through router



STAGE 10 — What you see in real time Terminal

4 (host2) shows your message arriving:

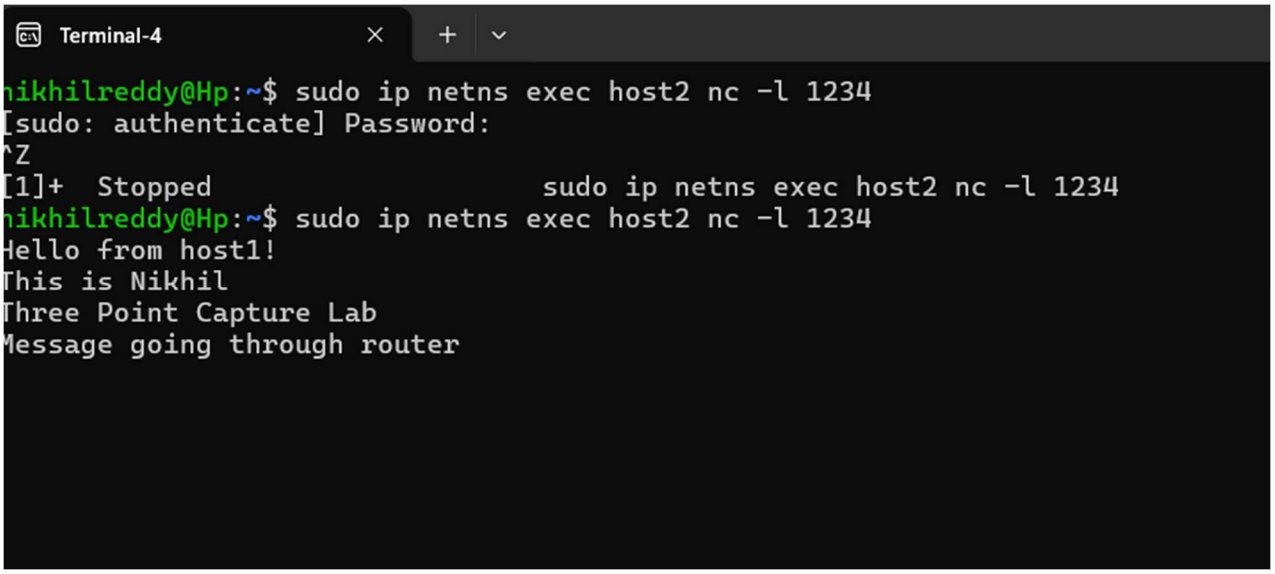
Hello from host1!

This is Alekhya

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Message travelling through router

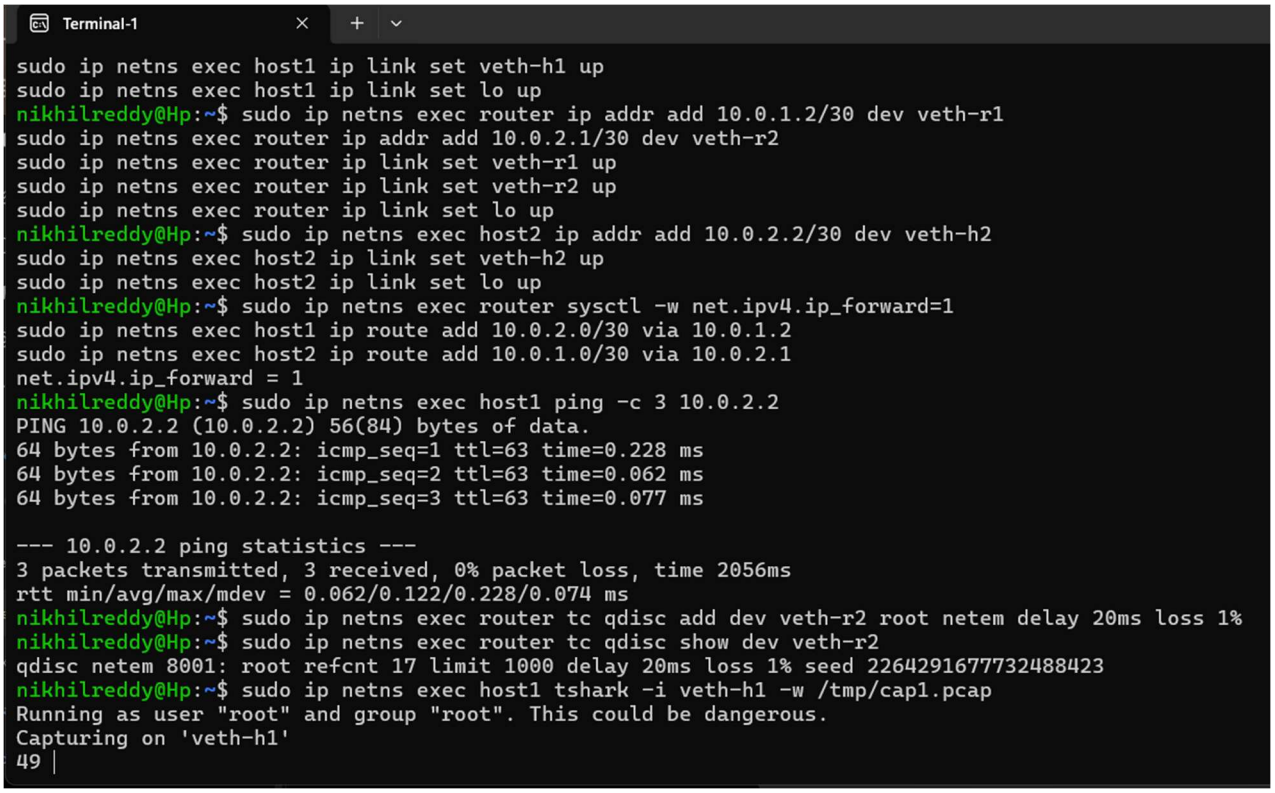


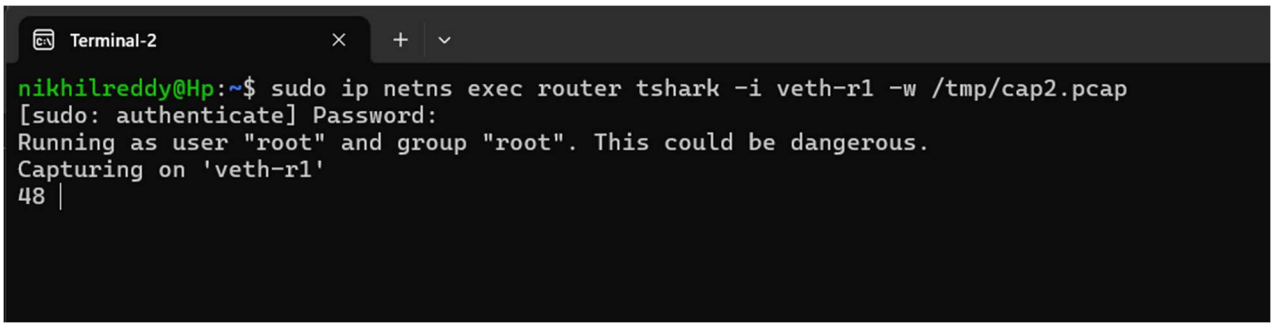
Terminal 1, 2, 3 (tshark) show packet counts increasing:

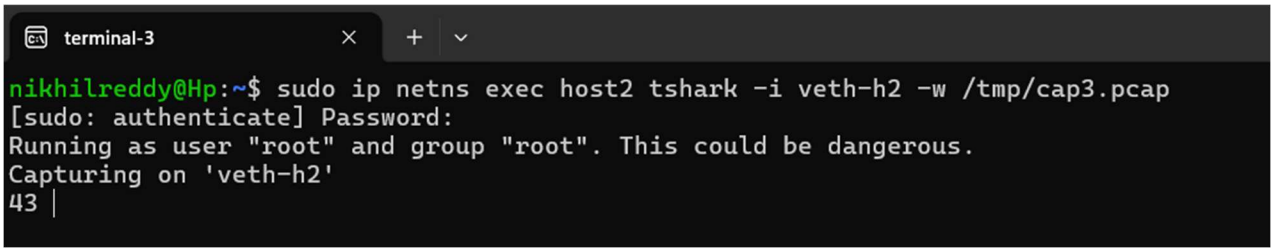
Capturing on 'veth-h1'

10 packets
24 packets

31 packets

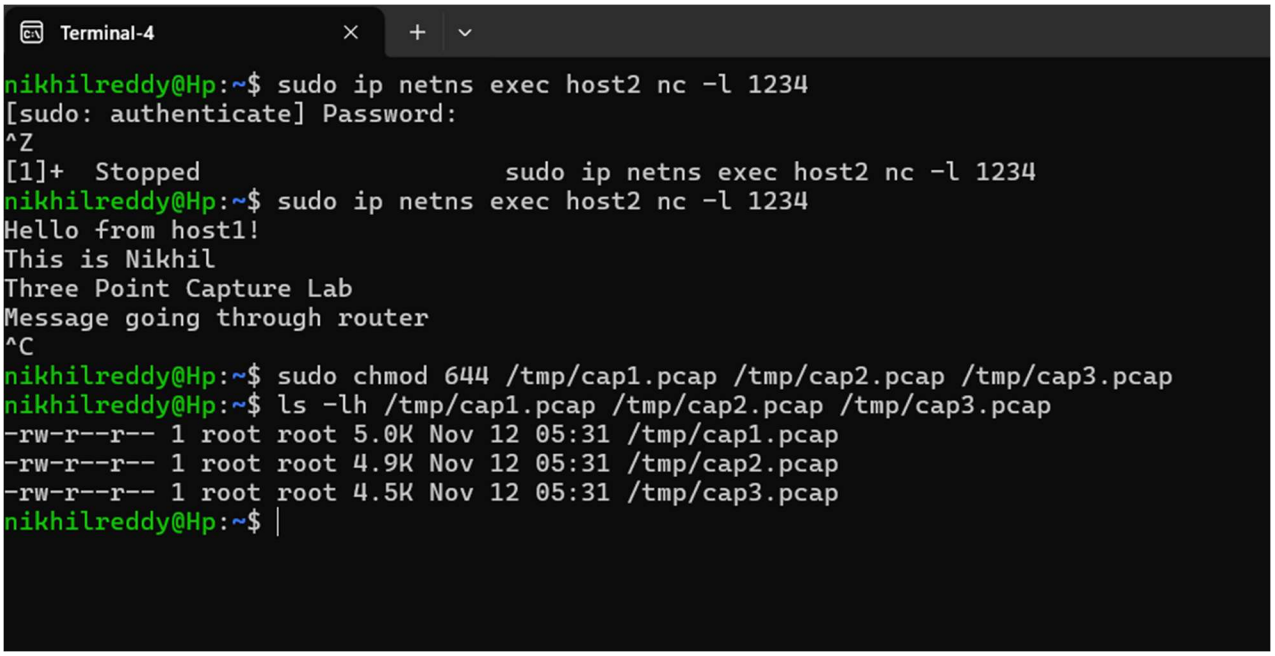






STAGE 12 — Fix permissions sudo chmod 644

/tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap



STAGE 13 — Check TTL

tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

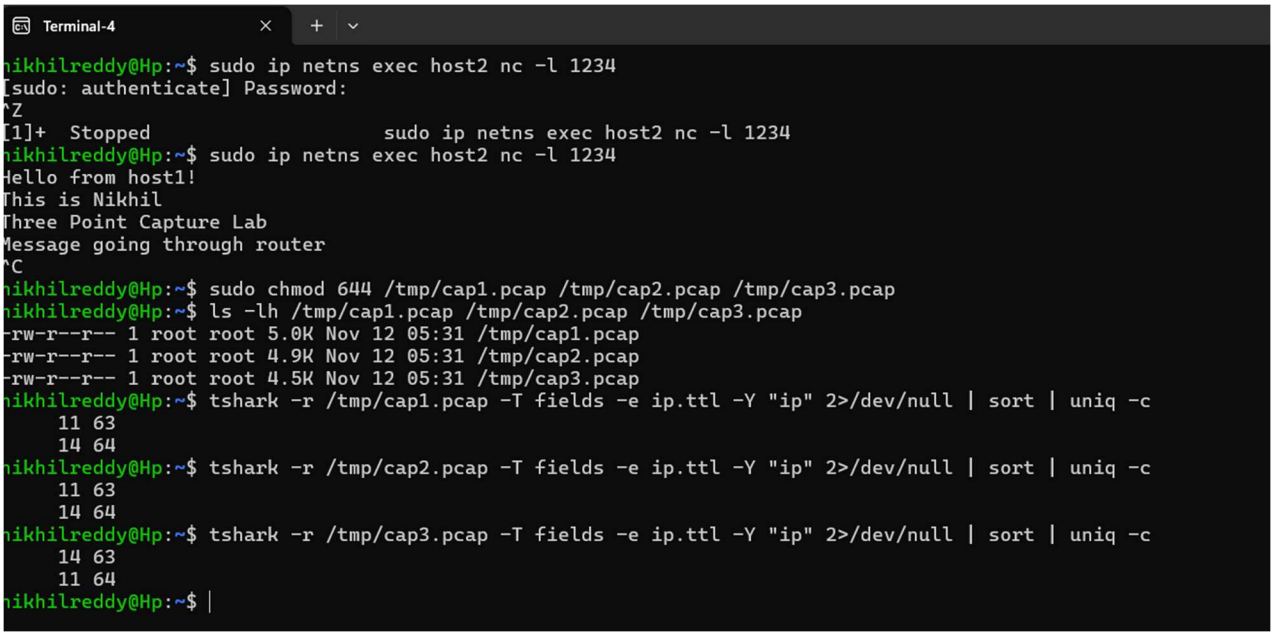
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

Expected: cap1

→ 64 cap2 →

64 cap3 → 63



STAGE 14 — Read your actual message from the capture

tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e data.text 2>/dev/null

You will see your actual message text extracted from the packet:

Hello from host1!

This is Alekhya

Computer Networks Lab

Three Point Capture Experiment

Message travelling through router



Connect from host1:

TERMINAL 1 — Setup (run all commands here)

Install tools:

sudo apt install -y tshark iproute2 netcat-openbsd wireshark

```
nikhilreddy@Hp:~$ sudo apt install -y tshark iproute2 netcat-openbsd wireshark
[sudo: authenticate] Password:
tshark is already the newest version (4.6.4-1).
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
iproute2 set to manually installed.
netcat-openbsd is already the newest version (1.234-1).
wireshark is already the newest version (4.6.4-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 168
nikhilreddy@Hp:~$
```

When asked → choose **Yes Create**

3 nodes:

sudo ip netns add host1 sudo
ip netns add router sudo ip
netns add host2

Verify:

sudo ip netns list

```
nikhilreddy@Hp:~$ sudo ip netns add host1 sudo ip netns add router sudo ip netns add host2
Cannot create namespace file "/run/netns/host1": File exists
nikhilreddy@Hp:~$ sudo ip netns list
host2 (id: 2)
router (id: 1)
host1 (id: 0)
nikhilreddy@Hp:~$
```

You must see: host1 router host2

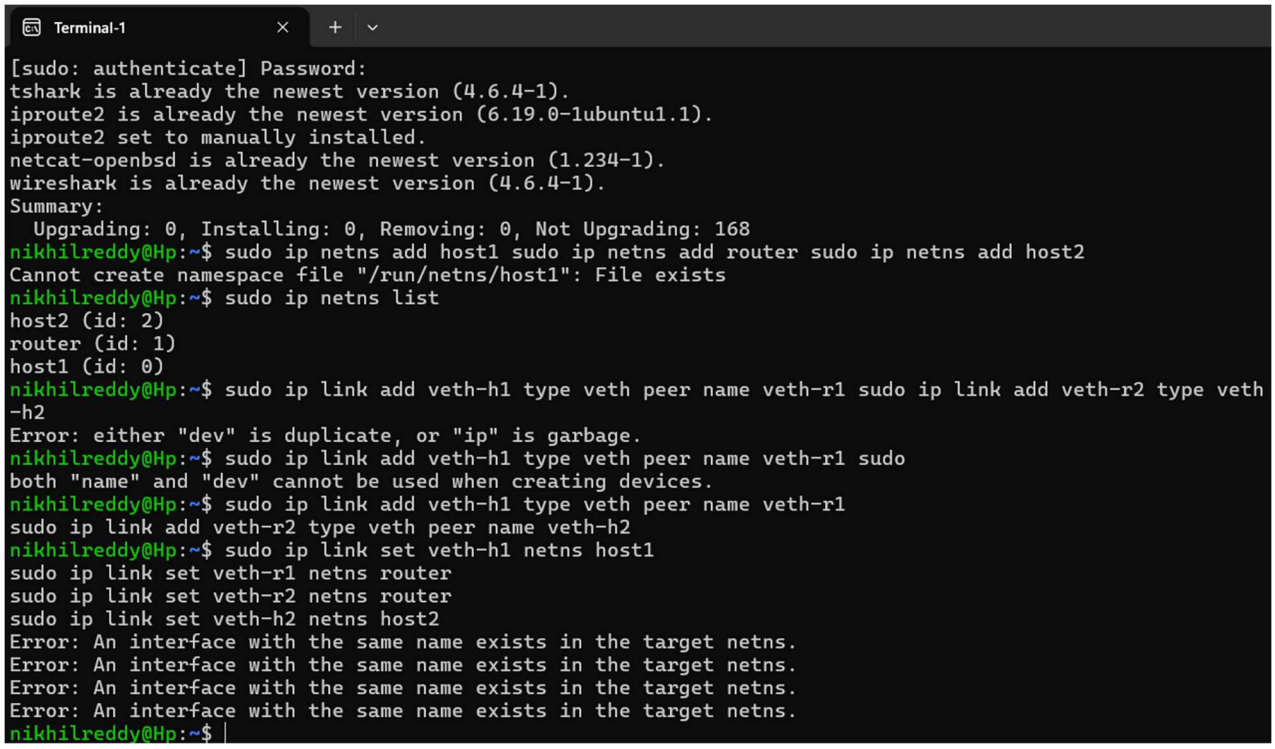
Create cables:

sudo ip link add veth-h1 type veth peer name veth-r1 sudo
ip link add veth-r2 type veth peer name veth-h2

```
nikhilreddy@Hp:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
nikhilreddy@Hp:~$
```

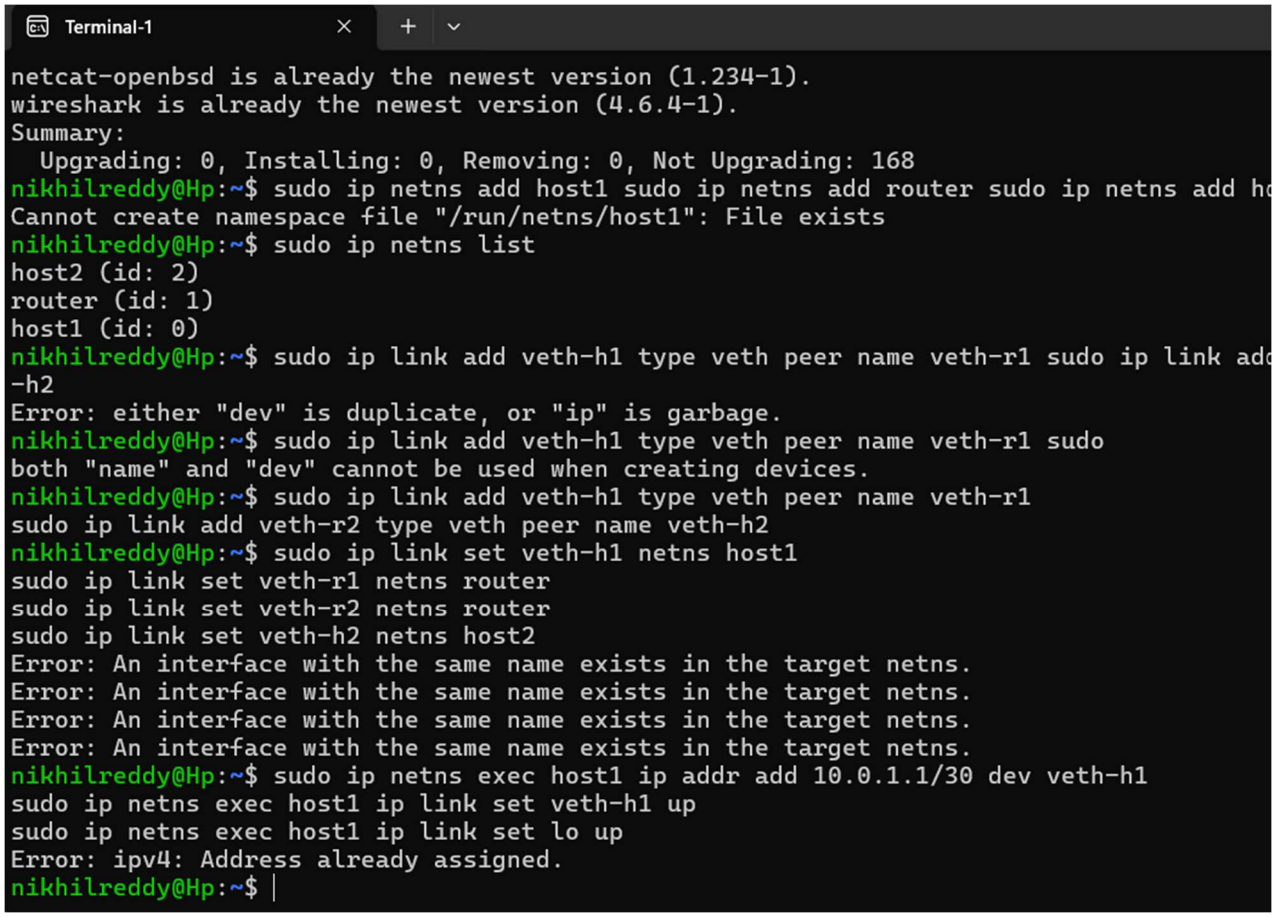
Plug cables into nodes:

sudo ip link set veth-h1 netns host1 sudo
ip link set veth-r1 netns router sudo ip
link set veth-r2 netns router sudo ip link
set veth-h2 netns host2



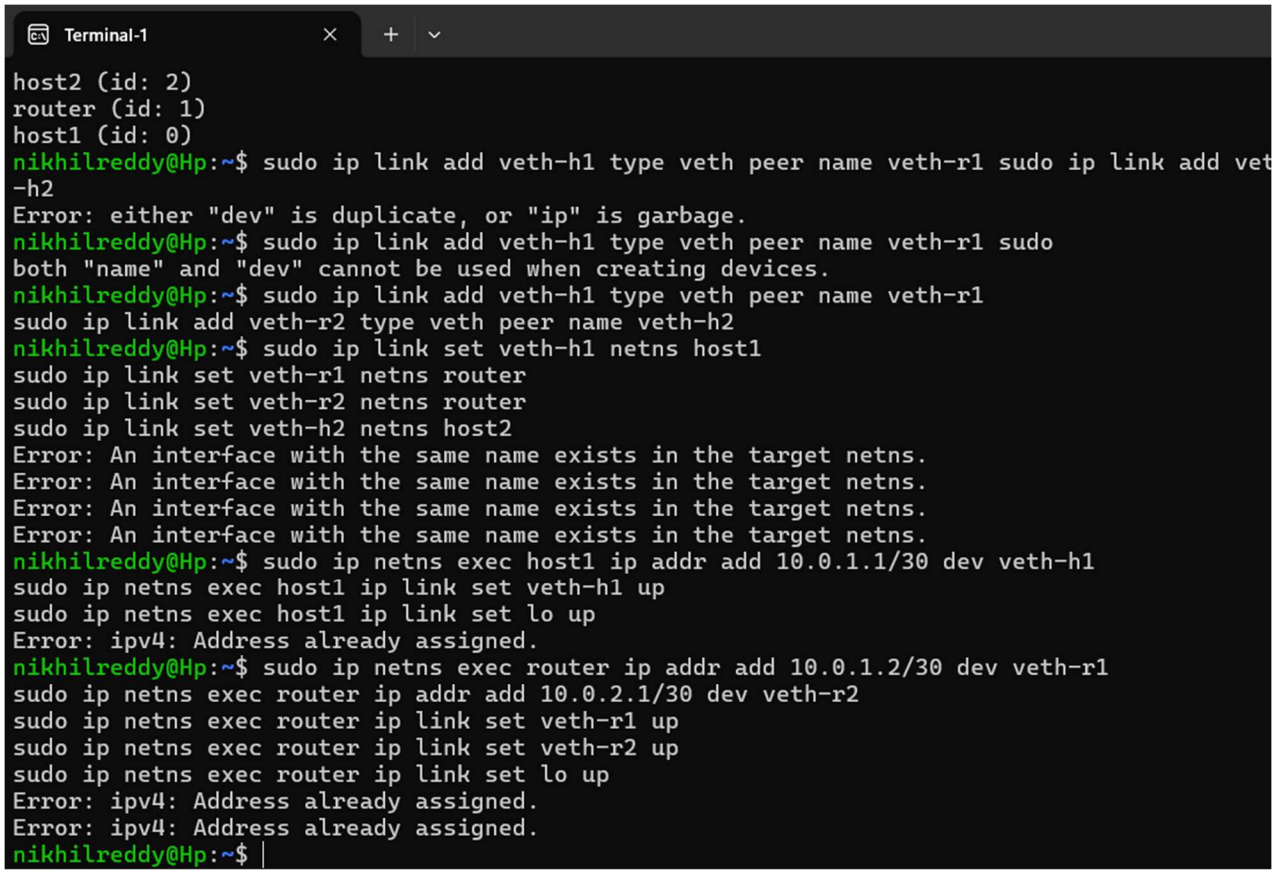
Configure host1:

sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up sudo ip netns
exec host1 ip link set lo up

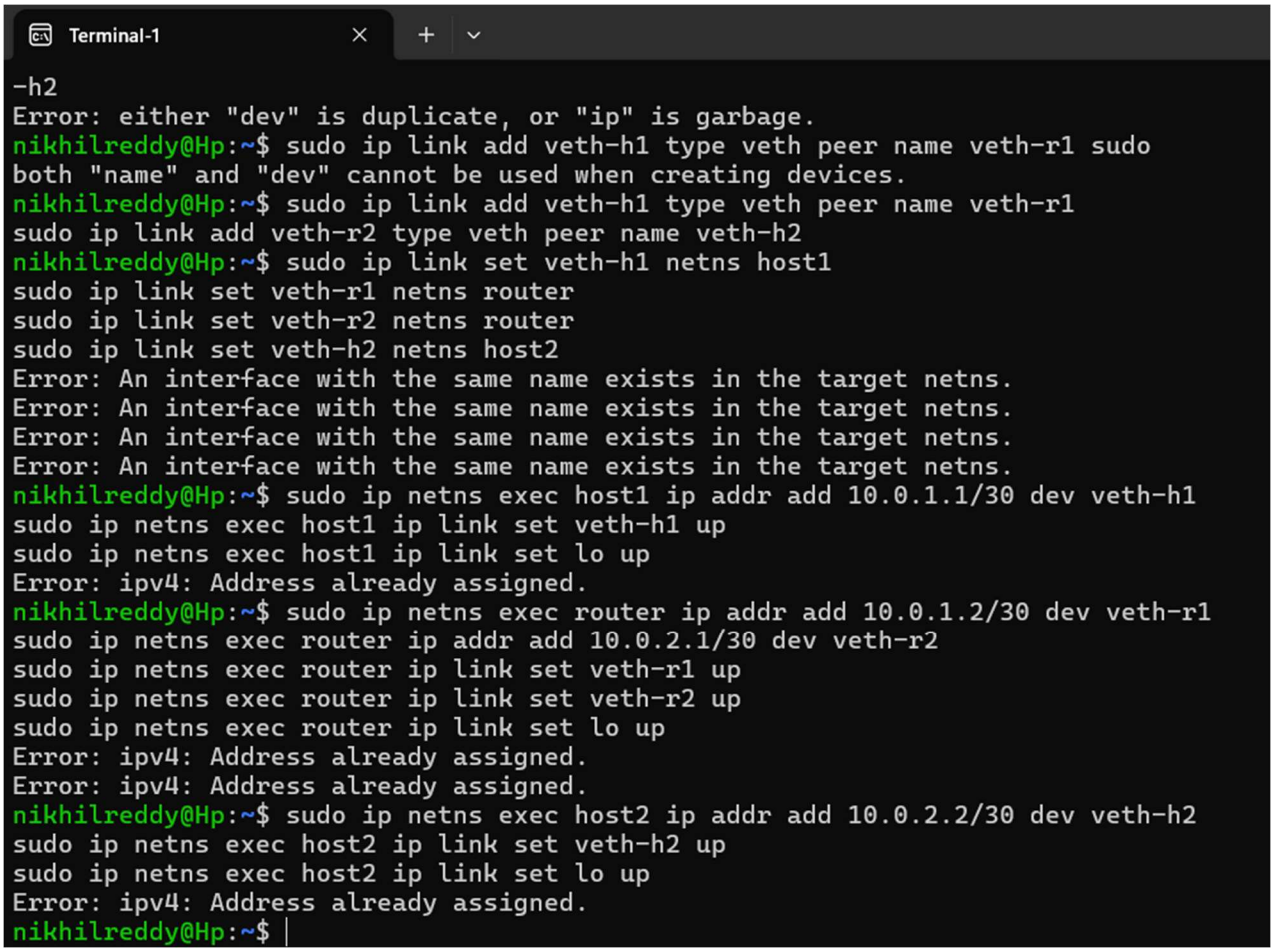


Configure router:

sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up sudo ip netns exec router ip link set veth-r2 up sudo ip netns exec router ip link set lo up

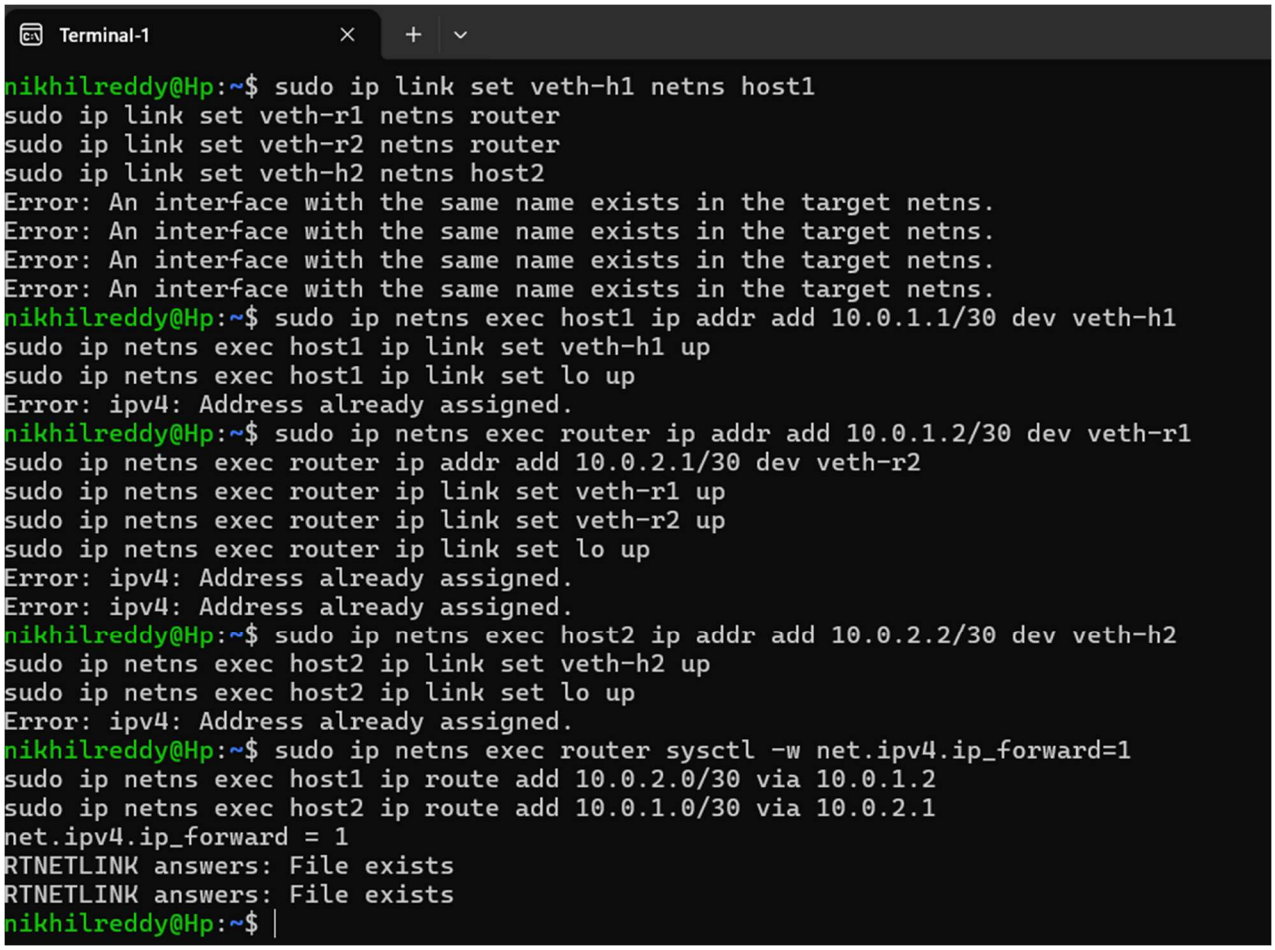


Configure host2: sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2 sudo ip netns exec host2 ip link set veth-h2 up sudo ip netns exec host2 ip link set lo up



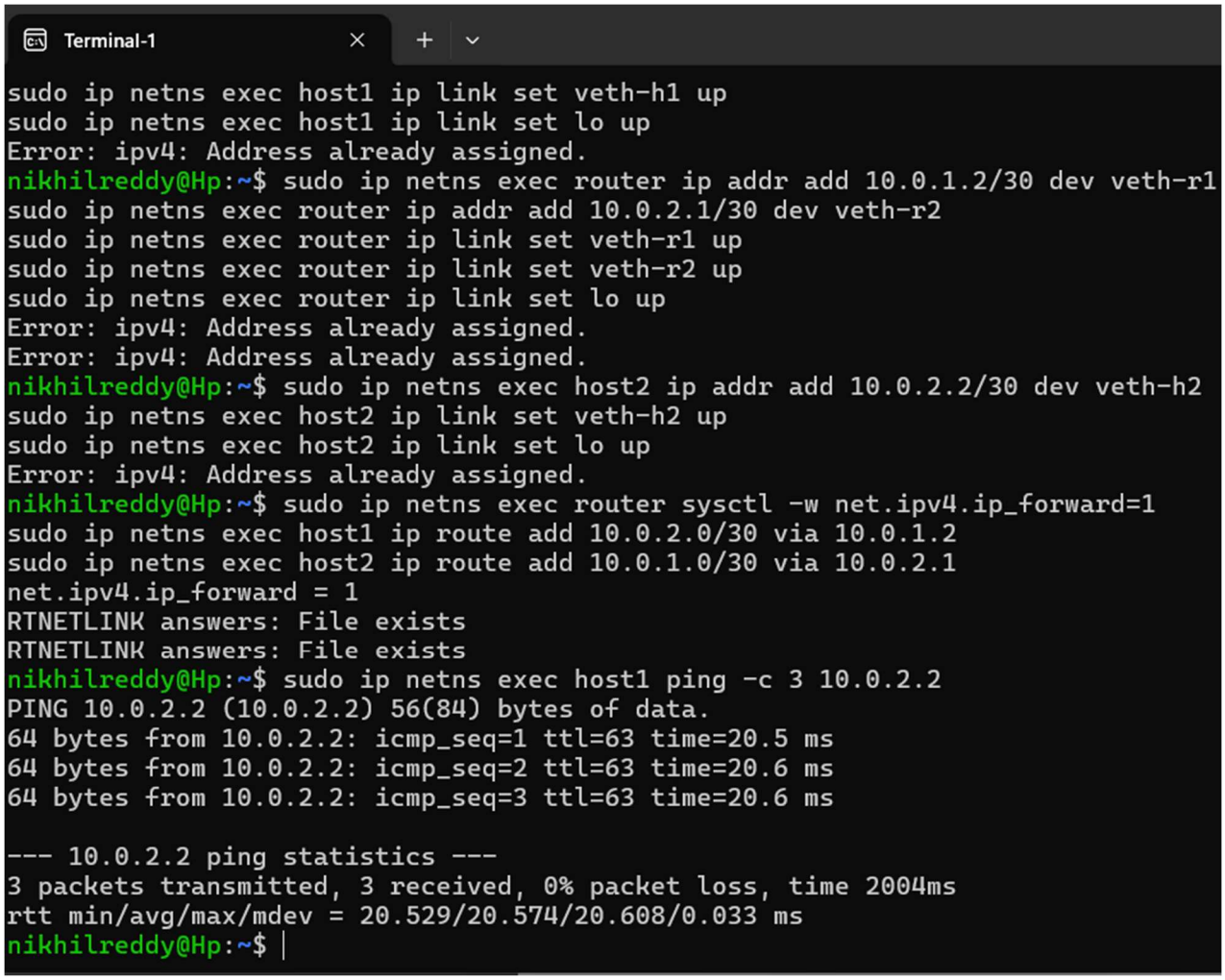
Enable routing:

sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1 sudo
ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2 sudo
ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1



Test connectivity:

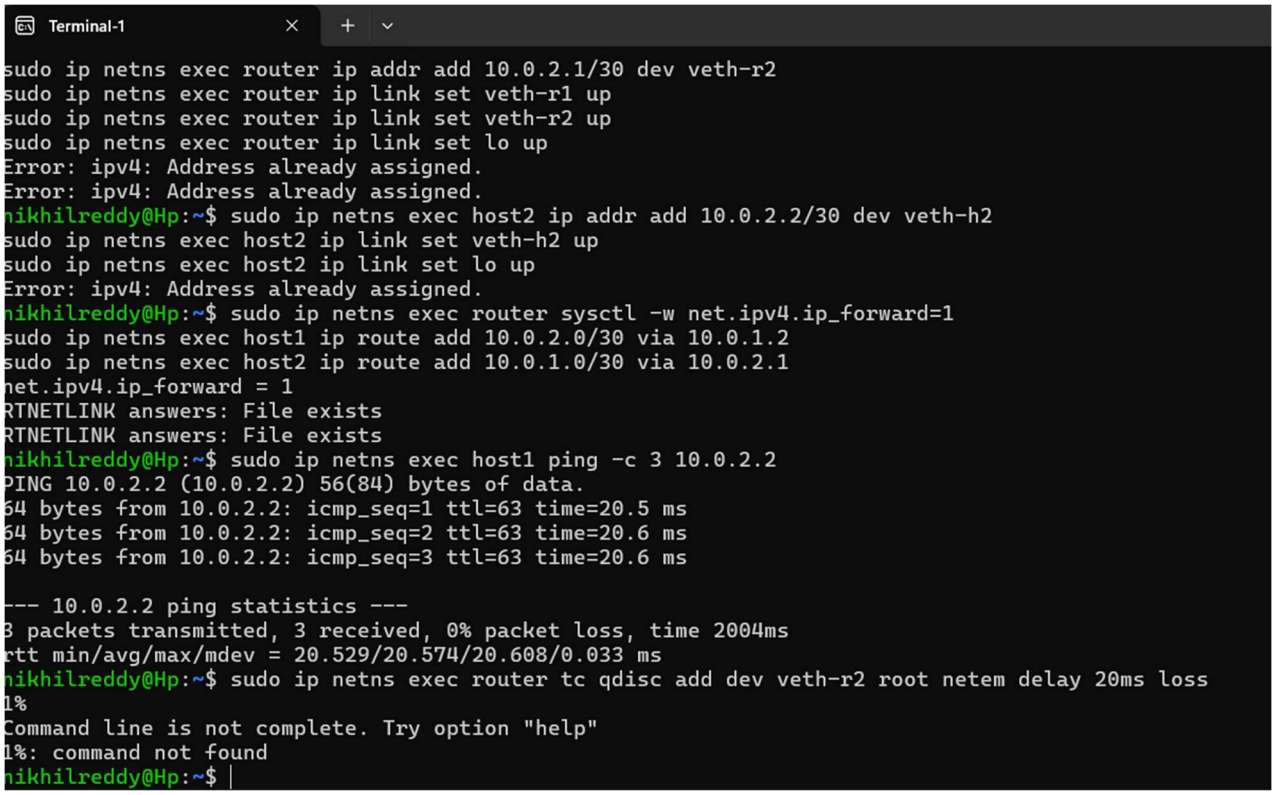
sudo ip netns exec host1 ping -c 3 10.0.2.2



Must see: 3 packets transmitted, 3 received, 0% packet loss

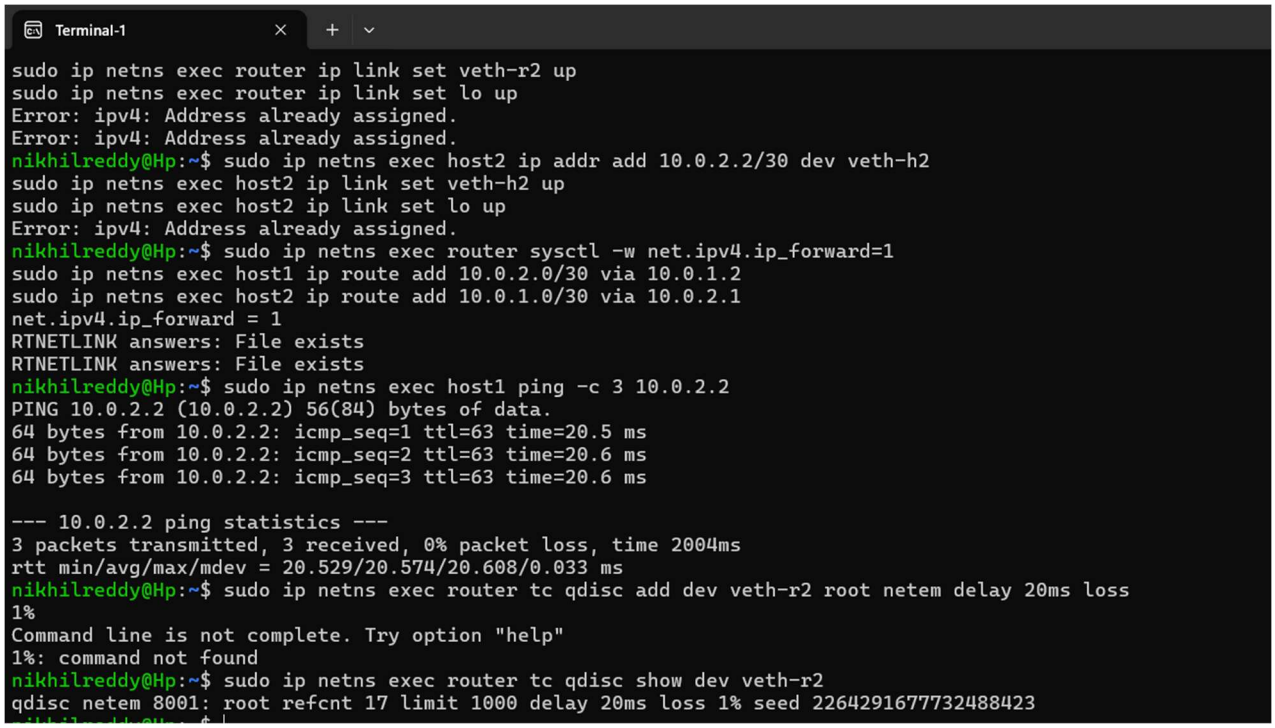
Add delay and loss:

sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%



Verify:

sudo ip netns exec router tc qdisc show dev veth-r2

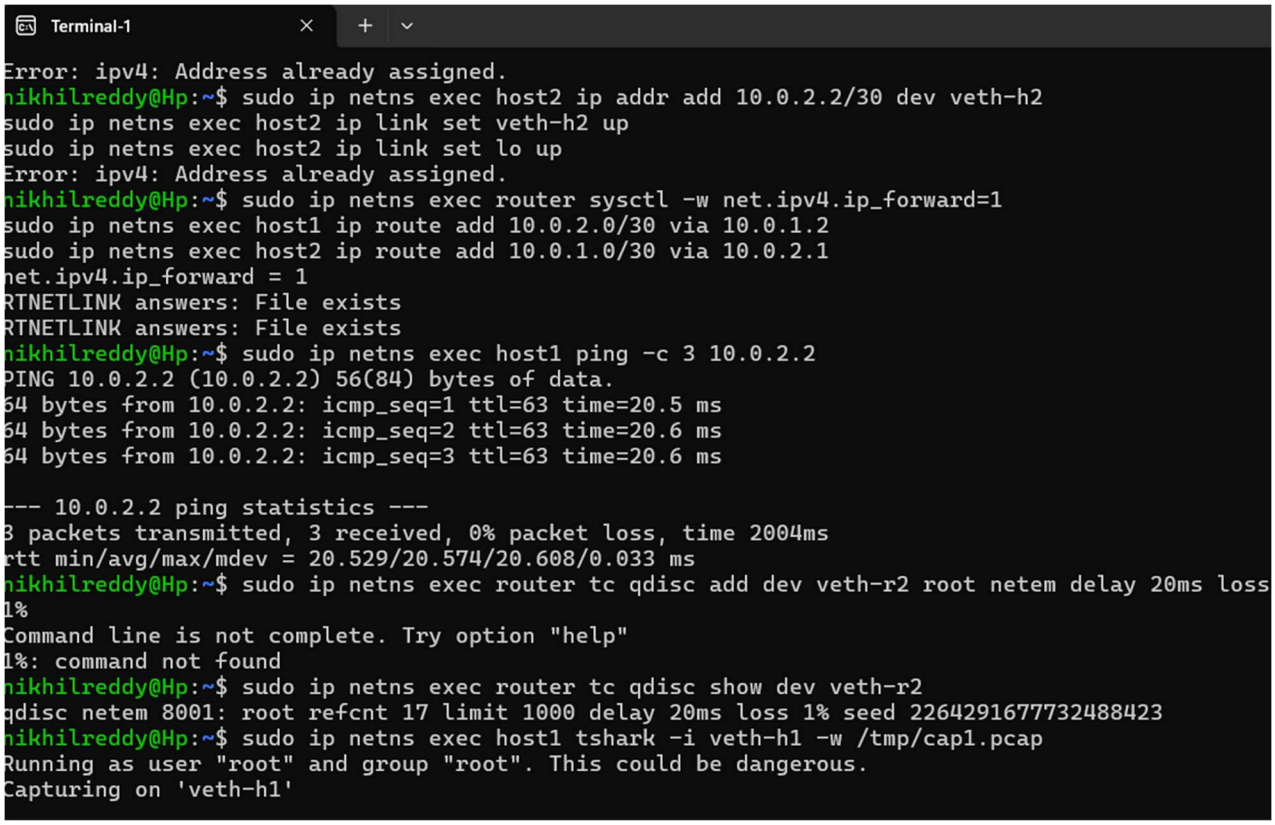


Must see: netem delay 20ms loss 1%

NOW USE ALL 4 TERMINALS

TERMINAL 1 — Capture at host1

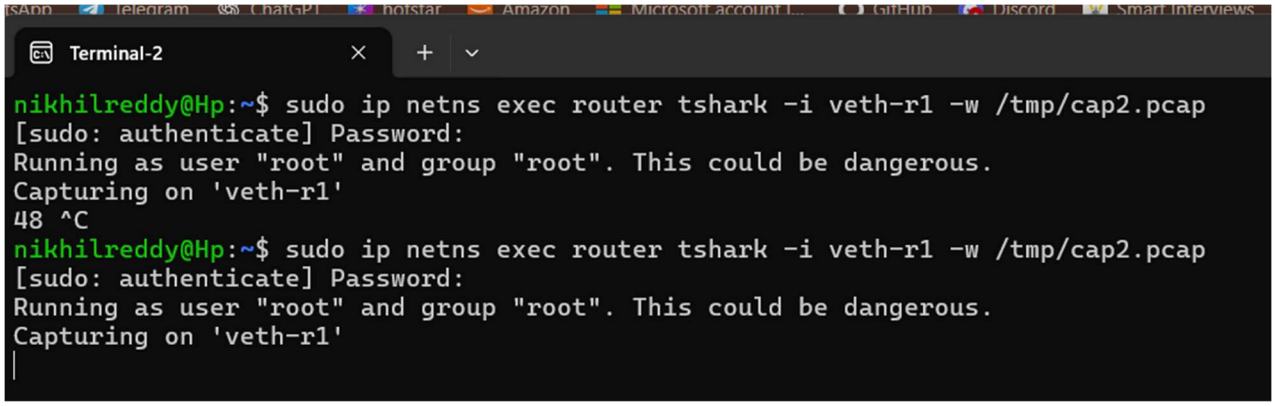
sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap



This terminal is left running.

TERMINAL 2 — Capture at router

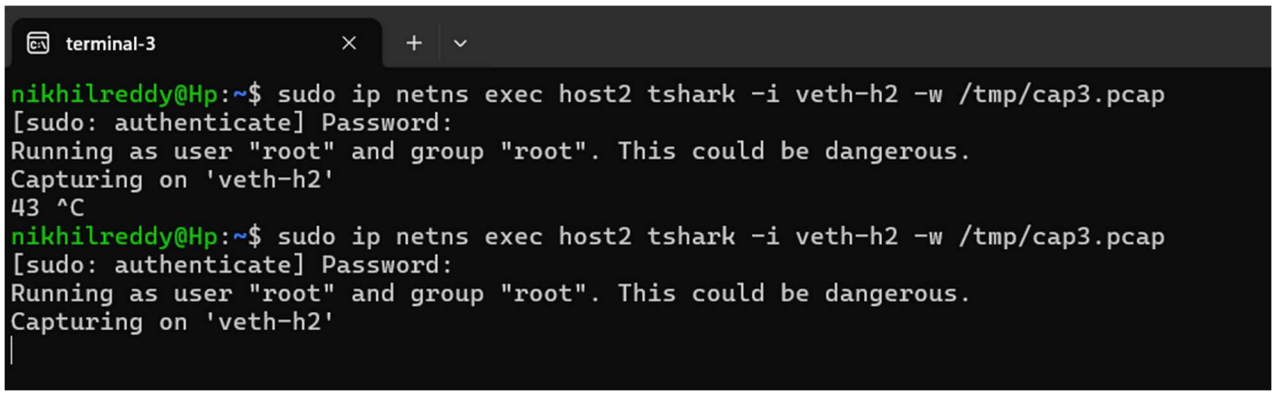
sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap



Leave running.

TERMINAL 3 — Capture at host2

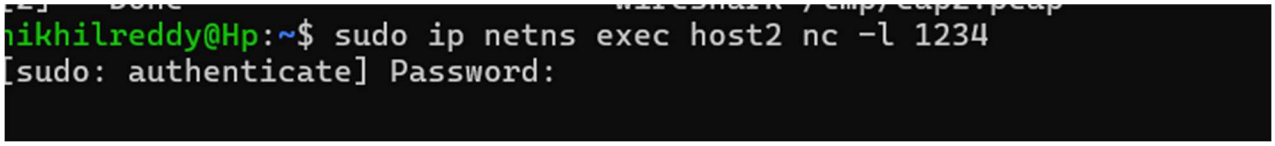
sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap



Leave running.

TERMINAL 4 — Send the message

Step 1 — Start receiver on host2:



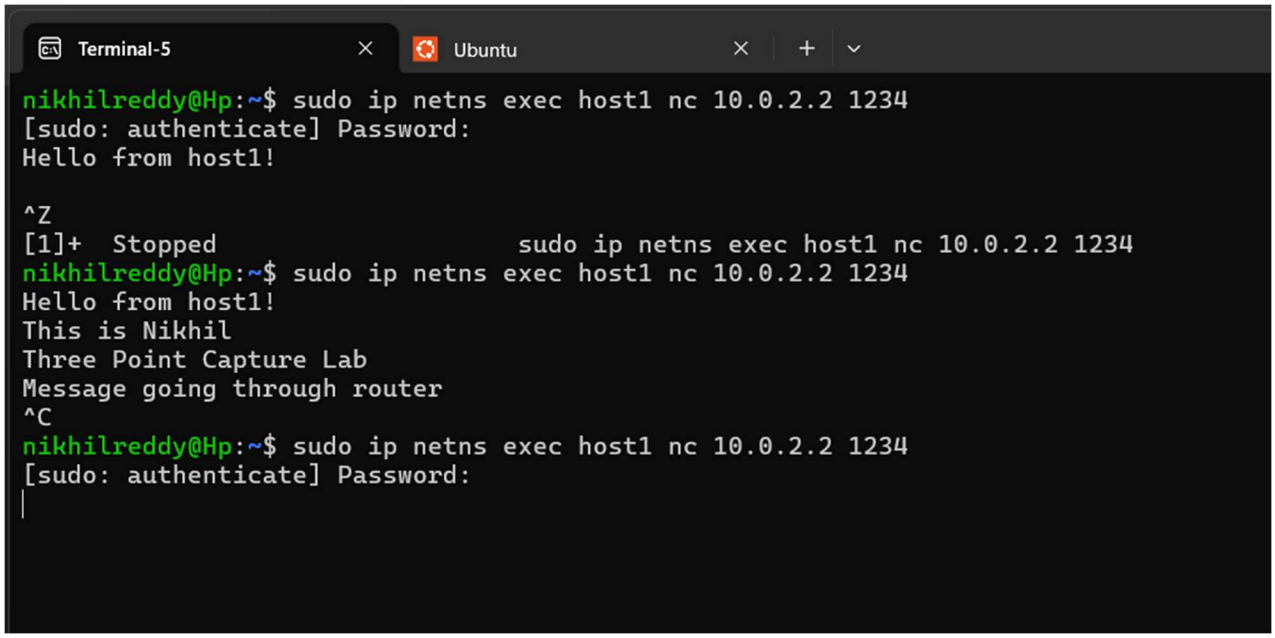
Cursor blinks. Leave it waiting.

Step 2 — Open a new Terminal 5

Step 3 — Connect from host1:

Paste the below command in terminal-5 sudo

ip netns exec host1 nc 10.0.2.2 1234



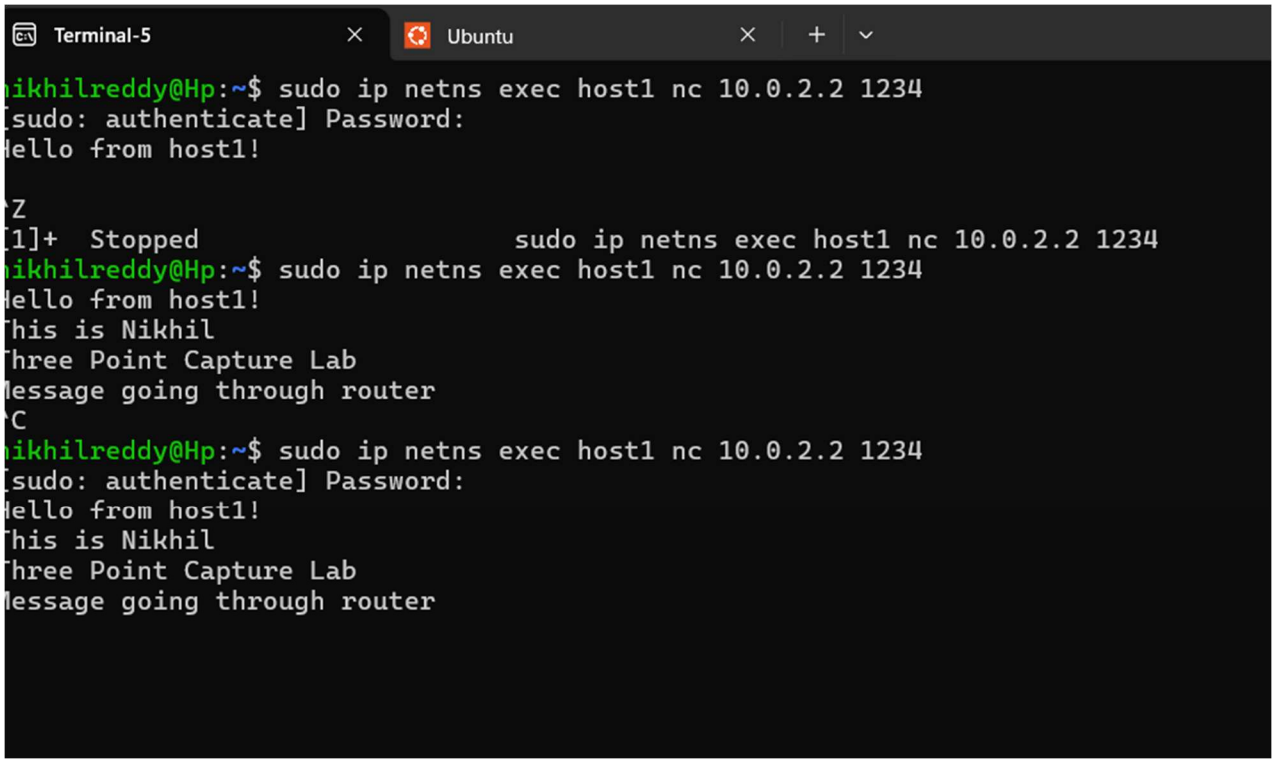
Type the below message in Terminal-5

Hello from host1!

This is Alekhya

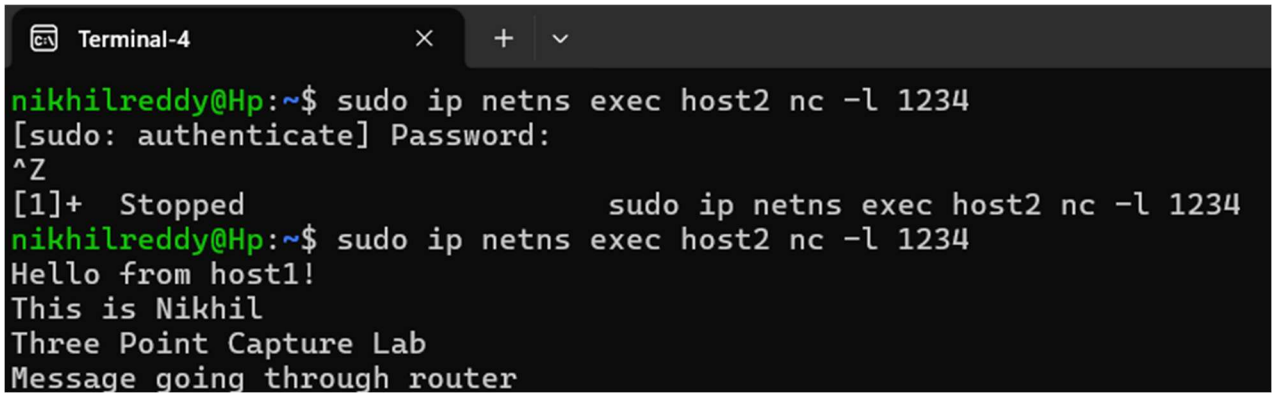
Three Point Capture Lab

Message going through router



Step 5 — Check Terminal 4 first tab

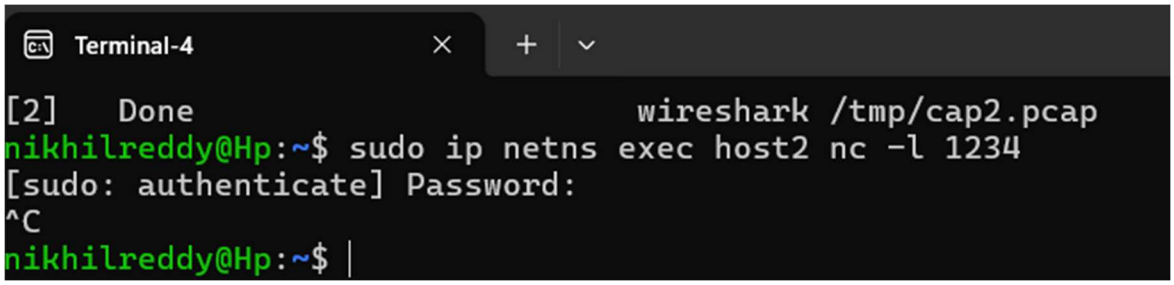
You will see your message appearing there — that is host2 receiving it.



Step 6 — Press Ctrl+C in all terminals

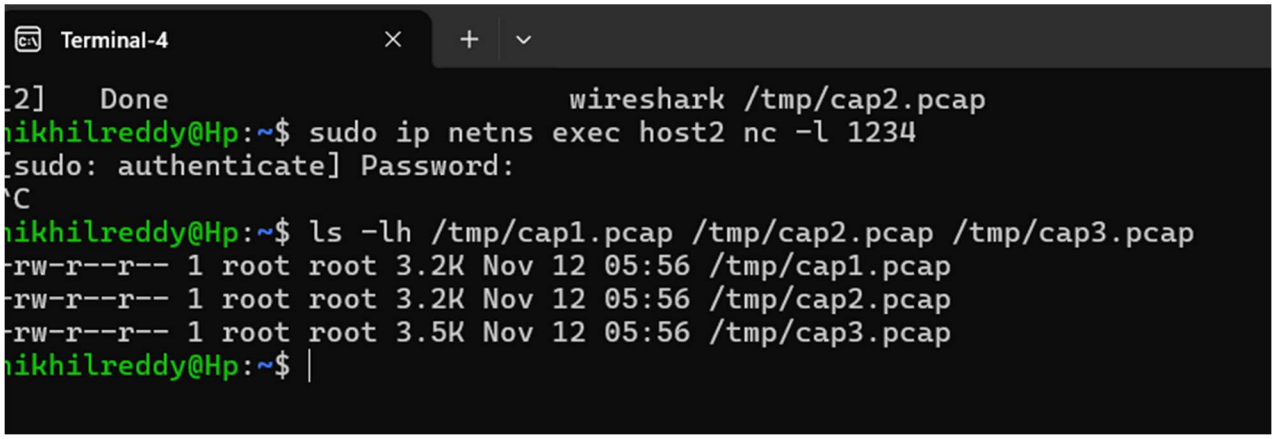
TERMINAL 4 — Analyse results Fix

permissions:



Check files are not empty:

ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap



All must show size greater than 0.

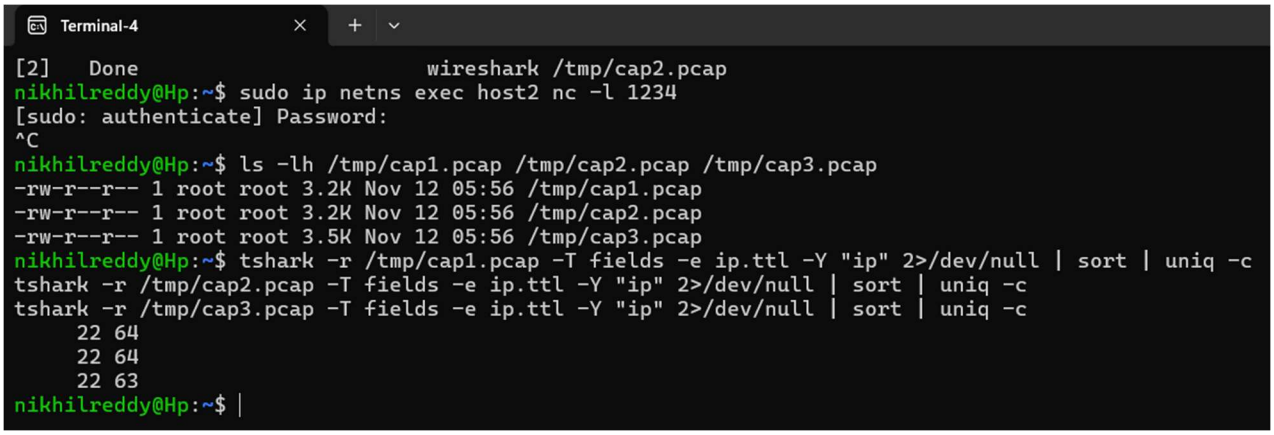
Check TTL at all 3 points:

tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c

Expected Outcome:



Read your actual message from router capture:

tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e tcp.payload 2>/dev/null | xxd | head -20

Check delay:

tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null

tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null

tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null Difference between cap1 and cap3 ≈ 20ms

Check retransmissions:

tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l

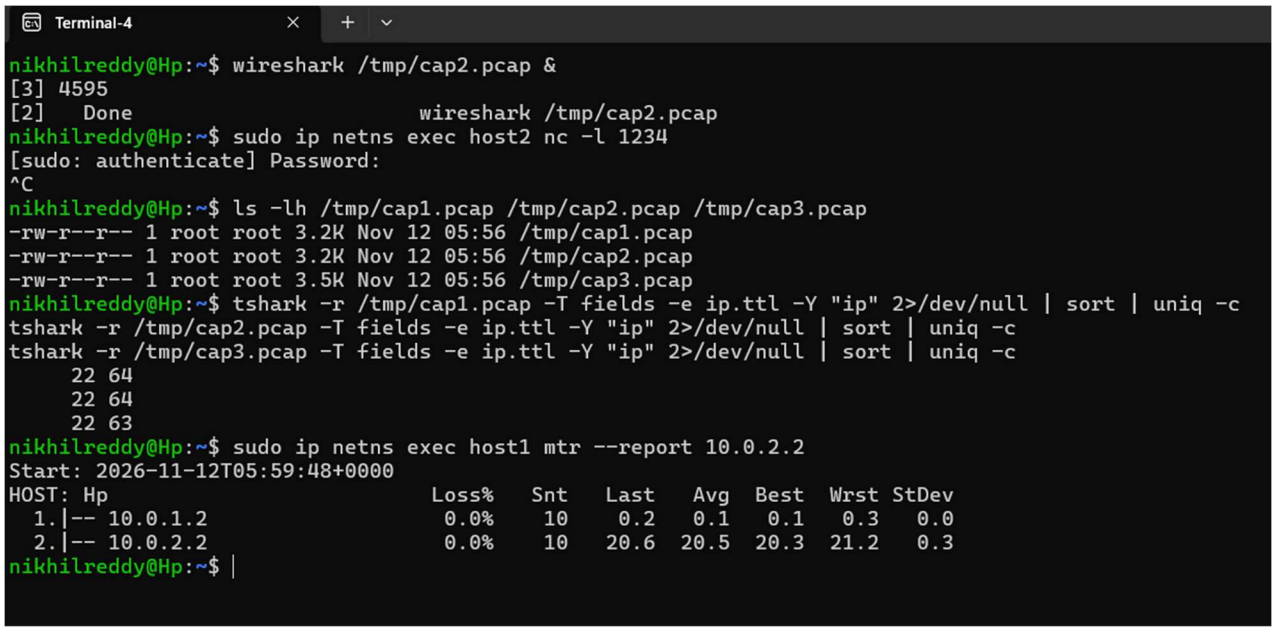
tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l

tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l

cap3 shows highest count — loss happened between router and host2.

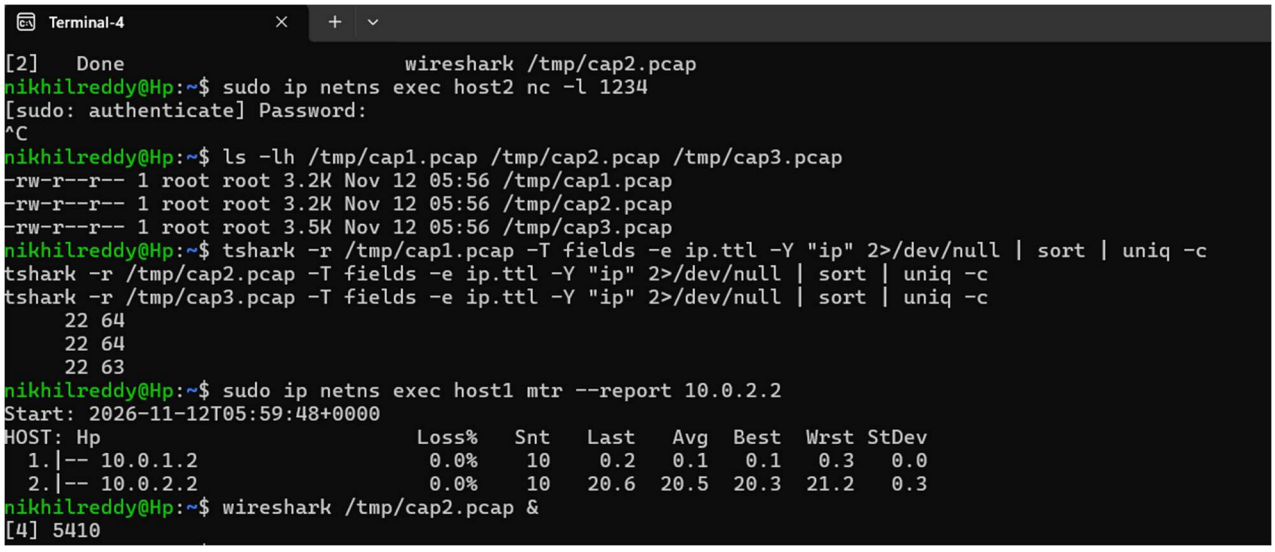
Check RTT:

sudo ip netns exec host1 mtr --report 10.0.2.2



TERMINAL 4 — Open Wireshark wireshark

/tmp/cap2.pcap &



Inside Wireshark: **Filter**

your message only:

tcp.port == 1234

Click any packet → expand middle pane → you will see:

▼ Ethernet II

▼ Internet Protocol Version 4

Time to Live: 64 ← TTL visible here

▼ Transmission Control Protocol
▼ Data

Hello from host1! ← your actual message visible here

Merge all 3 captures:

File → Merge → select cap1.pcap File

→ Merge → select cap3.pcap

Add TTL column:

Right-click column header

→ Column Preferences

→ click +

→ Title: TTL

→ Type: Custom

→ Fields: ip.ttl

→ OK

Students can now see TTL = 64 at points (1) and (2) and TTL = 63 at point (3)

Show packet loss:

Analyze → Expert Information CLEAN

UP — Terminal 4

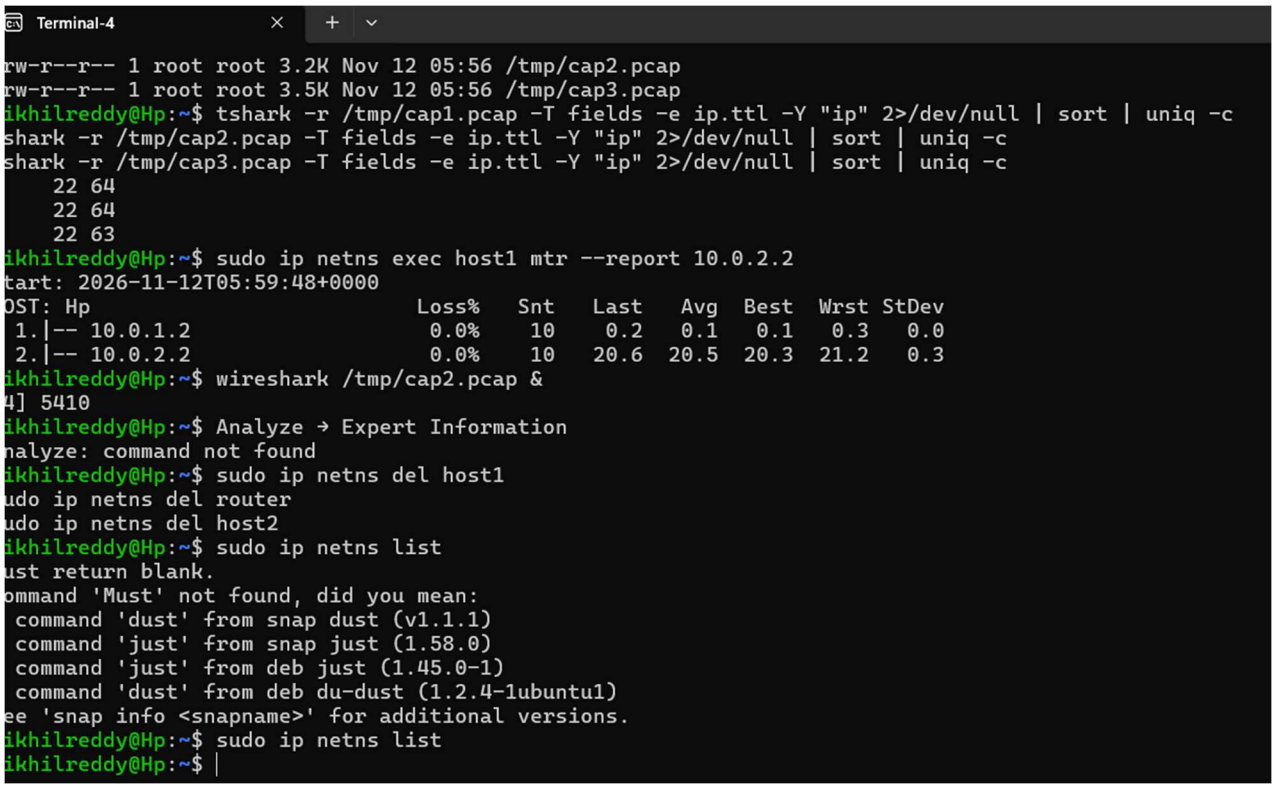
sudo ip netns del host1 sudo

ip netns del router sudo ip

netns del host2 **Verify:**

sudo ip netns list

Must return blank.


Terminal-4
rw-r--r-- 1 root root 3.2K Nov 12 05:56 /tmp/cap2.pcap
rw-r--r-- 1 root root 3.5K Nov 12 05:56 /tmp/cap3.pcap
ikhilreddy@Hp:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
shark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
shark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
22 64
22 64
22 63
ikhilreddy@Hp:~$ sudo ip netns exec host1 mtr --report 10.0.2.2
tart: 2026-11-12T05:59:48+0000
OST: Hp Loss% Snt Last Avg Best Wrst StDev
1.|-- 10.0.1.2 0.0% 10 0.2 0.1 0.1 0.3 0.0
2.|-- 10.0.2.2 0.0% 10 20.6 20.5 20.3 21.2 0.3
ikhilreddy@Hp:~$ wireshark /tmp/cap2.pcap &
4] 5410
ikhilreddy@Hp:~$ Analyze → Expert Information
nalyze: command not found
ikhilreddy@Hp:~$ sudo ip netns del host1
udo ip netns del router
udo ip netns del host2
ikhilreddy@Hp:~$ sudo ip netns list
ust return blank.
ommand 'Must' not found, did you mean:
command 'dust' from snap dust (v1.1.1)
command 'just' from snap just (1.58.0)
command 'just' from deb just (1.45.0-1)
command 'dust' from deb du-dust (1.2.4-1ubuntu1)
ee 'snap info <snapname>' for additional versions.
ikhilreddy@Hp:~$ sudo ip netns list
ikhilreddy@Hp:~$